Euler-Based Simulated Likelihood for Multidimensional Diffusions:

Dimensional Variance, Joint Asymptotics, and an Application Revisited

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Abstract

A widely used simulation method for likelihood inference in discretely observed diffusions approximates each transition density by simulating an Euler scheme to one step before the observed endpoint and averaging the final Gaussian density. This endpoint estimator is a shrinking-bandwidth kernel estimator whose Monte Carlo variance grows with the discretisation level at a dimension-dependent rate: in K dimensions the second moment of one simulated contribution is of order $M^{K/2}$, so the variance is of order $M^{K/2}/S$ rather than uniformly $1/S$. We derive exact Brownian moments, the corrected central-limit normalisation $\sqrt{S/M^{K/2}}$, the resulting bias–variance trade-off with optimal rate $S^{-4/(K+4)}$, and the scale of the nonlinear log-likelihood error. We then construct a direct counterexample to the joint transition-density convergence asserted in Lemma 2 of Brandt and Santa-Clara (2002): along $M = S \to \infty$, which satisfies their rate condition $\sqrt{S}/M \to 0$ and their own regularity assumptions, the simulated density converges in probability to zero for every $K > 2$ although the true density is strictly positive. Consequently the published proofs of Theorems 1 and 2 fail, and in four dimensions their rate condition, equivalent to $S/M^2 \to 0$, is incompatible with pointwise mean-square consistency of the simulated transition density, which requires $S/M^2 \to \infty$. We do not prove that the resulting simulated-likelihood argmax estimator is inconsistent, and we set out what such a proof would require. We then examine the mathematical status of the four-dimensional exchange-rate application: square-root boundaries, the reality of the incompleteness state along simulated paths, an implicitly defined volatility process whose reported loadings admit either zero or two positive branches, positive-semidefiniteness of the Brownian correlation matrix, a minimum-incompleteness selection rule with a corner solution, and binding inequality constraints. The paper does not assert that the published numerical estimates are incorrect; it establishes that their stated asymptotic justification is incomplete and, in the four-dimensional case, contradicted by an exact counterexample.

Keywords: simulated maximum likelihood; diffusion processes; Euler scheme; transition density; Monte Carlo variance; curse of dimensionality; diffusion bridges; nonregular inference.

JEL classifications: C13; C15; C32; G12; G15.

1 Introduction

Continuous-time diffusion models are often observed only at discrete dates. Their exact transition densities are rarely available, which makes exact maximum likelihood difficult even when simulation of approximate sample paths is straightforward. One influential response is the simulation method introduced by Pedersen (1995a) and Pedersen (1995b) and developed and applied by Brandt and Santa-Clara (2002). The method divides each observation interval into M Euler substeps, simulates the process through the first $M-1$ steps, evaluates the Gaussian density of the final Euler step at the observed endpoint, and averages this density over S simulated paths.

For fixed M, the construction is elementary: the sample average converges to the transition density of the M-step Euler chain as $S \to \infty$. For fixed S it is also clear that making the final time step smaller turns the Gaussian endpoint density into an increasingly narrow and increasingly high function. The mathematical question is therefore not whether the two separate approximations are individually meaningful. It is whether they remain compatible when M and S increase jointly.

The central observation of this paper is that the final Gaussian density is a kernel with bandwidth $h^{1/2}$, where $h = 1/M$. In K dimensions its effective volume is of order $h^{K/2}$ and its height is of order $h^{-K/2}$. Only simulations that fall in a shrinking neighbourhood of the endpoint contribute materially. The effective number of such simulations is therefore of order

$$Sh^{K/2} = \frac{S}{M^{K/2}}.$$

This quantity, rather than S alone, controls the Monte Carlo approximation.

That the endpoint construction is statistically wasteful is not a new observation. Durham and Gallant (2002) note that unconditioned Euler paths rarely arrive near the observed endpoint and propose bridge-based proposals for that reason; Elerian et al. (2001) augment the path with a Metropolis–Hastings block for the same reason; Stramer and Yan (2007) document numerically that the accuracy of the endpoint estimator deteriorates as M increases with S held fixed; and Lindström (2012) regularises the Pedersen proposal towards a bridge on the same diagnosis. What that literature records is a qualitative and largely numerical finding about efficiency. What is established here is different in kind: an exact dimensional law for the moments of the endpoint summand, the identification of $S/M^{K/2}$ as the quantity that governs the estimator, the corrected triangular-array normalisation, and a joint-limit counterexample showing that the published transition-density convergence statement is not merely unproved but false. The distinction matters because an efficiency loss can be absorbed by more simulations, whereas failure of the density estimator along an admissible rate path cannot.

The paper makes six main contributions.

First, we derive exact finite-M moments for the estimator under K-dimensional Brownian motion. The Euler approximation is exact in this benchmark, so any failure is caused by the endpoint Monte Carlo construction and not by discretisation bias, boundary behaviour, nonlinear coefficients, or parameter identification.

Second, we construct a direct counterexample to the joint transition-density convergence asserted in Lemma 2 of Brandt and Santa-Clara (2002). For every $K > 2$, choosing $M = S = n$ makes the simulated density at the Brownian origin converge in probability to zero. Standard Brownian motion satisfies the regularity assumptions of that paper exactly, and the sequence satisfies its condition $\sqrt{S}/M \to 0$, so the counterexample cannot be excluded by strengthening the hypotheses. Consequently the published proofs of Theorems 1 and 2 fail, since both rest on Lemma 2. In the four-dimensional application, their condition requires $S/M^2 \to 0$, while the variance calculation requires $S/M^2 \to \infty$ for mean-square consistency of the simulated transition density.

We are careful about what this does and does not establish. The counterexample concerns the simulated transition-density estimator at a point. It does not by itself exhibit a sequence along which the simulated-likelihood argmax estimator fails to converge, and we do not claim one. Section 1.1 states the logical status of each result, and Section 7.6 sets out what an argmax counterexample would require.

Third, we establish the generic local moment law for a smooth uniformly elliptic diffusion. If G_h denotes one endpoint-density summand, then for $r > 1$,

$$h^{(r-1)K/2}\mathbb{E}[G_h^r] \longrightarrow (2\pi)^{-(r-1)K/2} r^{-K/2} p(y \mid x)|V(y)|^{-(r-1)/2}.$$

In particular, $\mathrm{Var}(G_h)$ is of order $h^{-K/2}$. This gives the corrected triangular-array central-limit normalisation

$$\sqrt{Sh^{K/2}}\,(\widehat{q}_{M,S} - q_M),$$

not $\sqrt{S}(\widehat{q}_{M,S} - q_M)$.

Fourth, we combine the $O(M^{-1})$ Euler density bias used by Brandt and Santa-Clara (2002) with the corrected simulation variance. The resulting pointwise mean squared error is

$$O(M^{-2}) + O\left(\frac{M^{K/2}}{S}\right),$$

which is minimised at $M \asymp S^{2/(K+4)}$ and yields the rate $S^{-4/(K+4)}$. The estimator therefore exhibits an explicit curse of dimensionality.

Fifth, we trace the density-level result into the log likelihood. The nonlinearity of the logarithm operates on the effective Monte Carlo size $S/M^{K/2}$. Whenever a standard second-order delta expansion is valid, the leading log-density bias is of order $M^{K/2}/S$, rather than uniformly of order $1/S$. This distinction matters because simulated likelihood sums a log transformation over all observations.

Sixth, we examine the mathematical status of the exchange-rate application. We first give finite-sample scaling diagnostics showing that the ratios corresponding to the paper's own rate conditions are not small at the implemented design, so the implementation does not resemble a point along a sequence in which those conditions operate. The empirical model then contains square-root diffusions that violate the smoothness and nondegeneracy assumptions of the generic theorem, though we show that for the interest rates this violation is numerically inconsequential at the reported estimates. The market-incompleteness state is not: its Feller ratio equals one half for every parameter value, and because the estimator eliminates that state in favour of volatility, every Euler substep requires a recovered square root to remain real. The time-varying risk-price specification defines exchange-rate volatility through an implicit quadratic identity whose leading coefficient is negative at the reported U.S.–U.K. estimates, so that system admits either zero or two positive volatility branches and never exactly one. The full Brownian correlation matrix requires a joint positive-semidefinite constraint, not merely pairwise correlations in $[-1, 1]$. Finally, the minimum-incompleteness rule is a linear programme whose solution is always a corner, which we verify against the published tables, and together with the binding nonnegativity constraint it yields a hybrid and potentially nonregular estimator not covered by the unconstrained SML theorem.

The aim is not to relitigate the substantive economics of a paper published more than two decades ago. Nor do we infer from a false theorem that every numerical estimate obtained with finite M and S must be poor. The aim is narrower and mathematical: to identify the correct dimensional scale of the endpoint estimator, separate sequential from joint convergence, and determine what the published asymptotic arguments do and do not establish.

1.1 Scope of the results

Because this paper both proves theorems and reassesses a published application, it matters which claims are which. Every result below falls into one of three categories, and the text signals the category wherever a result is stated.

Proved. The exact Brownian moments of Proposition 3.1, the collapse theorem Theorem 3.3, the general local moment law Theorem 4.2, the density variance scale of Corollary 4.6, the density-level triangular-array central limit theorems Theorem 3.7 and Theorem 4.7, and the pointwise mean-squared-error balance of Proposition 5.1. These are theorems about the simulated *transition density*, proved under stated assumptions.

Conditional. The log-density expansion Proposition 6.1, which requires uniform integrability and lower-tail control that we assume rather than verify; the resulting statements about scores and Hessians; and the abstract perturbation result Proposition 6.2, which is a deterministic implication and not a theorem about the simulated estimator.

Diagnostic. Everything in Section 8 concerning the exchange-rate model: the finite-sample scaling ratios, the square-root boundary behaviour, correlation-matrix feasibility, the volatility state transformation, minimum-incompleteness geometry, and constrained inference. These are calculations about a specific published model, not asymptotic theorems, and they are reported as such.

One boundary deserves emphasis because it is easy to blur. All the proved results concern the simulated transition density. None of them concerns the maximiser of the simulated log likelihood. We do not prove that the endpoint simulated-likelihood estimator is inconsistent, and Section 7.6 and Section B set out precisely what such a proof would require.

The remainder is organised as follows. Section 2 defines the estimator. Section 3 gives exact Brownian results and the counterexample. Section 4 derives the general diffusion moment law and corrected CLT. Section 5 studies the bias–variance trade-off. Section 6 discusses log likelihood and parameter estimation. Section 7 analyses the published proof. Section 8 studies the exchange-rate diffusion. Section 9 presents reproducible numerical evidence, and Section 10 discusses variance-reduction alternatives.

2 The endpoint simulated-likelihood estimator

Let $Y_t \in \mathbb{R}^K$ satisfy the stochastic differential equation

$$dY_t = \mu(Y_t, t; \theta)\, dt + \Sigma(Y_t, t; \theta)\, dW_t, \qquad V = \Sigma\Sigma^\mathsf{T}, \tag{1}$$

where W_t is a Brownian motion of suitable dimension and $\theta \in \boldsymbol{\Theta} \subset \mathbb{R}^L$. Coefficients are assumed regular enough for a unique strong solution in the sense of Karatzas and Shreve (1991). We suppress t and θ where no ambiguity arises. Consider one observation interval of length one, from $Y_0 = x$ to $Y_1 = y$. The normalisation to unit length is only notational.

For $M \geq 2$, set $h = 1/M$ and define the Euler chain (Kloeden and Platen, 1992)

$$Y_{m+1}^M = Y_m^M + \mu(Y_m^M, mh)h + \Sigma(Y_m^M, mh)\sqrt{h}\,\varepsilon_{m+1}, \qquad m = 0, \ldots, M-1, \tag{2}$$

with $Y_0^M = x$ and independent $\varepsilon_m \sim \mathcal{N}(0, I)$. Conditional on $Y_m^M = z$, the one-step density is

$$g_h(y \mid z) = \varphi_K(y; z + \mu(z, 1-h)h, hV(z, 1-h)). \tag{3}$$

Let $Z_h = Y_{M-1}^M$ denote the preterminal Euler state and let $r_h(z \mid x)$ be its density. The M-step Euler transition density is

$$q_M(y \mid x) = \mathbb{E}[g_h(y \mid Z_h)] = \int_{\mathbb{R}^K} g_h(y \mid z) r_h(z \mid x)\, dz. \tag{4}$$

The endpoint simulation estimator is

$$\widehat{q}_{M,S}(y \mid x) = \frac{1}{S} \sum_{s=1}^{S} g_h(y \mid Z_{h,s}), \tag{5}$$

where $Z_{h,1}, \dots, Z_{h,S}$ are independent draws from the preterminal Euler law. This is equation (8) of Brandt and Santa-Clara (2002), with notation adapted to the present analysis.

For each fixed M, the strong law yields

$$\widehat{q}_{M,S}(y \mid x) \longrightarrow q_M(y \mid x) \quad \text{almost surely as } S \to \infty,$$

provided the summand is integrable. Under regularity conditions such as those of Bally and Talay (1995) and Bally and Talay (1996),

$$q_M(y \mid x) - p(y \mid x) = O(M^{-1}),$$

where p is the diffusion transition density. These two facts establish a sequential limit: first $S \to \infty$ for fixed M, then $M \to \infty$. They do not establish convergence along arbitrary joint sequences $M, S \to \infty$.

The geometry of (3) explains why. Its covariance is $hV(z)$. Under nondegeneracy, it is concentrated in a ball of radius $O(\sqrt{h})$ around the endpoint and has maximum height $O(h^{-K/2})$. The integral in (4) remains finite because the effective volume is $O(h^{K/2})$. A Monte Carlo average resolves that integral only if enough preterminal draws enter this shrinking region.

Definition 2.1 (Effective endpoint simulation size)**.** For a K-dimensional endpoint estimator with Euler step $h = 1/M$, define

$$R_{M,S} = Sh^{K/2} = \frac{S}{M^{K/2}}. \tag{6}$$

This is the expected-order number of simulations falling in an $O(\sqrt{h})$ neighbourhood of an endpoint with a regular positive preterminal density.

The next sections make this interpretation exact.

3 Exact Brownian analysis

Consider standard Brownian motion in $\mathbb{R}^K$,

$$dY_t = dW_t, \tag{7}$$

observed over a unit interval. Euler discretisation is exact. This benchmark is not an adversarial construction outside the scope of the published theory: with $\mu \equiv 0$ and $\Sigma \equiv I_K$, all coefficient derivatives vanish and are therefore continuous and bounded of every order, so Assumption 1 of Brandt and Santa-Clara (2002) holds, and $\Sigma\Sigma^\intercal = I_K$ is positive definite, so Assumption 2 holds. Lemmas 1–3 of that paper are stated under exactly these two assumptions. Any failure exhibited here is therefore a failure inside the stated hypotheses, and cannot be repaired by strengthening the regularity conditions. Set $Y_0 = x$ and evaluate the transition density at $Y_1 = y$. With $h = 1/M$,

$$Z_h \sim \mathcal{N}(x, (1-h)I_K)$$

and a single endpoint summand is

$$G_h = (2\pi h)^{-K/2} \exp\left[-\frac{\|y - Z_h\|^2}{2h}\right]. \tag{8}$$

3.1 Exact moments

Proposition 3.1 (Exact Brownian moments)**.** *Let $r > 0$. For the summand* (8)*,*

$$\mathbb{E}[G_h^r] = (2\pi h)^{-rK/2}\left(1 + \frac{r(1-h)}{h}\right)^{-K/2} \exp\left[-\frac{r\|y-x\|^2}{2\{h + r(1-h)\}}\right] \tag{9}$$
$$= (2\pi)^{-rK/2} h^{-(r-1)K/2}\{r - (r-1)h\}^{-K/2} \exp\left[-\frac{r\|y-x\|^2}{2\{r-(r-1)h\}}\right].$$

In particular,

$$\mathbb{E}[G_h] = (2\pi)^{-K/2} \exp\left[-\frac{\|y-x\|^2}{2}\right] = p(y \mid x), \tag{10}$$

and

$$\mathbb{E}[G_h^2] = (2\pi)^{-K} h^{-K/2}(2-h)^{-K/2} \exp\left[-\frac{\|y-x\|^2}{2-h}\right]. \tag{11}$$

Proof. For $Z \sim \mathcal{N}(x, (1-h)I_K)$ and $c > 0$, the Gaussian identity

$$\mathbb{E}\left[e^{-c\|Z-y\|^2}\right] = \{1 + 2c(1-h)\}^{-K/2} \exp\left[-\frac{c\|x-y\|^2}{1+2c(1-h)}\right]$$

follows by completing the square. Apply it with $c = r/(2h)$ and multiply by the prefactor $(2\pi h)^{-rK/2}$. The cases $r = 1$ and $r = 2$ follow directly. □

Corollary 3.2 (Exact variance)**.** *For S independent simulations,*

$$\mathrm{Var}(\widehat{q}_{M,S}) = \frac{1}{S}\left[(2\pi)^{-K} h^{-K/2}(2-h)^{-K/2} e^{-\|y-x\|^2/(2-h)} - p(y \mid x)^2\right]. \tag{12}$$

Hence

$$\mathrm{Var}(\widehat{q}_{M,S}) \sim \frac{\tau_K^2(x,y)}{S h^{K/2}}, \qquad \tau_K^2(x,y) = (2\pi)^{-K} 2^{-K/2} e^{-\|y-x\|^2/2}. \tag{13}$$

The variance is not uniformly $O(1/S)$. It is $O(M^{K/2}/S)$. The estimator is unbiased in this example for every M, but its distribution becomes increasingly dominated by rare, very large contributions when $R_{M,S}$ is small.

3.2 A direct joint-limit counterexample

A variance calculation alone does not prove failure of convergence in probability because a non-uniformly integrable sequence can converge in probability while retaining large or divergent moments. The next theorem gives a direct failure result. It is a statement about the simulated transition density at a single pair of points; Section 7.6 explains why it does not transfer automatically to the parameter estimator.

Theorem 3.3 (Collapse along $M = S$)**.** *Let $K > 2$, $x = y = 0$, and choose $M_n = S_n = n$. Then the Brownian endpoint estimator satisfies*

$$\widehat{q}_{n,n}(0 \mid 0) \longrightarrow 0 \qquad \textit{in probability}, \tag{14}$$

although

$$p(0 \mid 0) = (2\pi)^{-K/2} > 0.$$

In every dimension the sequence satisfies the rate condition

$$\frac{\sqrt{S_n}}{M_n} = n^{-1/2} \longrightarrow 0$$

used in Theorems 1 and 2 of Brandt and Santa-Clara (2002), and it satisfies without any rate restriction the hypotheses of their Lemma 2, which requires only $M \to \infty$ and $S \to \infty$.

Proof. Write $h_n = 1/n$ and let $Z_{n,1}, \ldots, Z_{n,n}$ be the preterminal draws, each distributed as $\mathcal{N}(0, (1-1/n)I_K)$. Fix a constant $c > K$ and define

$$r_n = \sqrt{\frac{c \log n}{n}}.$$

Let A_n be the event that at least one draw lies in the ball $B(0, r_n)$. The Gaussian densities of the $Z_{n,s}$ are uniformly bounded for $n \geq 2$, so for a constant C_K,

$$\mathbb{P}(A_n) \leq n\mathbb{P}(\|Z_{n,1}\| \leq r_n) \leq C_K n r_n^K = C_K c^{K/2} n^{1-K/2} (\log n)^{K/2} \longrightarrow 0$$

because $K > 2$.

On A_n^c, every summand is bounded by

$$\left(\frac{n}{2\pi}\right)^{K/2} \exp\left(-\frac{n r_n^2}{2}\right) = (2\pi)^{-K/2} n^{(K-c)/2},$$

which tends to zero because $c > K$. The sample average obeys the same bound. Therefore, for every $\varepsilon > 0$,

$$\mathbb{P}(\widehat{q}_{n,n} > \varepsilon) \leq \mathbb{P}(A_n) + \mathbf{1}\left\{(2\pi)^{-K/2} n^{(K-c)/2} > \varepsilon\right\} \longrightarrow 0.$$

□

Remark 3.4 (How an unbiased estimator converges to zero)**.** For every n, $\mathbb{E}[\widehat{q}_{n,n}] = p(0 \mid 0)$. The convergence in probability to zero is sustained by increasingly rare simulations with increasingly large endpoint densities. The sequence is not uniformly integrable. This is precisely the rare-event geometry hidden by a fixed-M application of the strong law.

Corollary 3.5 (Contradiction in four dimensions)**.** *For $K = 4$, mean-square consistency requires*

$$\frac{S}{M^2} \longrightarrow \infty,$$

whereas the published condition $\sqrt{S}/M \to 0$ is equivalent to

$$\frac{S}{M^2} \longrightarrow 0.$$

The two conditions are mutually exclusive. Moreover, Theorem 3.3 shows that the published condition permits a sequence along which the density estimator converges to the wrong value.

Remark 3.6 (Where the conflict is unconditional)**.** The incompatibility in Corollary 3.5 is a property of dimensions $K \geq 4$. Mean-square consistency requires $S/M^{K/2} \to \infty$ and the published condition requires $S/M^2 \to 0$; for $K \geq 4$ the first demands $S \gtrsim M^{K/2} \geq M^2$ and the second demands $S \ll M^2$, so no sequence satisfies both. For $K \leq 3$ the two are compatible: with $K = 3$ and $S = M^{7/4}$ one has $S/M^{3/2} \to \infty$ and $\sqrt{S}/M \to 0$ simultaneously. Theorem 3.3 is nevertheless not confined to $K \geq 4$. It shows that even when the published condition admits consistent sequences, as at $K = 3$, it also admits inconsistent ones, so the condition cannot be what makes the theorem work. The empirical application of Brandt and Santa-Clara (2002) has $K = 4$, where no admissible sequence exists at all.

3.3 Correct triangular-array central limit theorem

Theorem 3.7 (Brownian endpoint CLT). *Let $M \to \infty$ and $S \to \infty$ such that*

$$Sh^{K/2} = \frac{S}{M^{K/2}} \longrightarrow \infty. \tag{15}$$

Then

$$\sqrt{Sh^{K/2}}\,\{\widehat{q}_{M,S}(y \mid x) - p(y \mid x)\} \Longrightarrow \mathcal{N}\Big(0, \tau_K^2(x,y)\Big), \tag{16}$$

where $\tau_K^2(x,y)$ is defined in (13)*.*

Proof. Index the array by n, with $h_n = 1/M_n$ and $s = 1, \dots, S_n$; within row n the summands $G_{n,s}$ are independent and identically distributed. Proposition 3.1 at $r = 2$ gives $h_n^{K/2}\mathbb{E}[G_{n,1}^2] \to \tau_K^2$, and $\mathbb{E}G_{n,1} = p(y \mid x)$ exactly for every M, so

$$h_n^{K/2}\operatorname{Var}(G_{n,1}) = h_n^{K/2}\mathbb{E}[G_{n,1}^2] - h_n^{K/2}p(y \mid x)^2 \longrightarrow \tau_K^2.$$

At $r = 3$ the same proposition gives $\mathbb{E}[G_{n,1}^3] = O(h_n^{-K})$, a bound on the *raw* moment. Since $G_{n,1} \geq 0$ and $\mathbb{E}G_{n,1} = p(y \mid x)$ is constant in n, the inequality $|a-b|^3 \leq 4(a^3 + b^3)$ for $a, b \geq 0$ gives the centred bound

$$\mathbb{E}|G_{n,1} - \mathbb{E}G_{n,1}|^3 \leq 4\left\{\mathbb{E}[G_{n,1}^3] + p(y \mid x)^3\right\} = O(h_n^{-K}).$$

The Lyapunov ratio at order three is therefore

$$\frac{S_n\mathbb{E}|G_{n,1} - \mathbb{E}G_{n,1}|^3}{\{S_n \operatorname{Var}(G_{n,1})\}^{3/2}} = \frac{S_n O(h_n^{-K})}{S_n^{3/2}\tau_K^3 h_n^{-3K/4}\{1 + o(1)\}} = O\Big(\big(S_n h_n^{K/2}\big)^{-1/2}\Big) \longrightarrow 0$$

by (15). The triangular-array Lyapunov central limit theorem applies. Because the Euler scheme is exact for Brownian motion, $q_M = p$ for every M and no bias term appears, so the statement may be centred at p directly. □

The correct normalisation depends on dimension. A fixed-M $\sqrt{S}$ central limit theorem is valid for each M separately, but it is not stable as $M \to \infty$: see Remark 4.8.

4 General multidimensional diffusions

The Brownian calculation reflects a local property of Gaussian endpoint kernels. We now derive the generic moment law. We state it under a clean global framework, so that every step of the proof is checkable and no condition is phrased in terms of the conclusion it is meant to deliver; Remark 4.4 explains how localisation would weaken it.

Throughout this section the model is time homogeneous, so that $\mu(z)$ and $V(z)$ carry no time argument. The one-step kernel (3) is then evaluated at the preterminal state alone, and no ambiguity arises about whether coefficients are taken at time $1-h$ or at time 1. Remark 4.5 records what changes otherwise.

Let $G_h = g_h(y \mid Z_h)$ be one summand in (5), where Z_h is the Euler chain at time $1-h$. Fix x and y.

Assumption 4.1 (Global regularity)**.***(A1)* *$\mu : \mathbb{R}^K \to \mathbb{R}^K$ is bounded and continuous, with $\|\mu\|_\infty = \bar{\mu} < \infty$.*

(A2) $V : \mathbb{R}^K \to \mathbb{R}^{K\times K}$ *is continuous and globally uniformly elliptic: there exist constants* $0 < \underline{v} \le \overline{v} < \infty$ *with*

$$\underline{v}\, I_K \preceq V(z) \preceq \overline{v}\, I_K \qquad \textit{for every } z \in \mathbb{R}^K. \tag{17}$$

(A3) *For each* $h \in (0, \frac{1}{2}]$ *the preterminal Euler state* Z_h*, that is the chain* (2) *at time* $1-h$*, has a density* $r_h(\cdot \mid x)$ *with respect to Lebesgue measure.*

(A4) *The family is uniformly bounded:* $\sup_{0<h\le 1/2}\ \sup_{z\in\mathbb{R}^K} r_h(z \mid x) = \bar{r} < \infty$.

(A5) *There is a continuous density* $p(\cdot \mid x)$ *with* $p(y \mid x) > 0$ *such that* $r_h(\cdot \mid x) \to p(\cdot \mid x)$ *uniformly on some closed ball* $\overline{B}(y, \delta_0)$*,* $\delta_0 > 0$*, as* $h \downarrow 0$*.*

Two comments on the content of these conditions. (A5) is a joint requirement in time and space: the preterminal state lives at time $1-h$, not at time 1, so the stated convergence combines convergence of the Euler law to the diffusion law with continuity of $t \mapsto p_t(\cdot \mid x)$ at $t = 1$ in the topology of uniform convergence near y. The limit $p(\cdot \mid x)$ is the diffusion transition density at time 1. And (A4) is not implied by (A3): it is a uniform-in-h statement, which for uniformly elliptic coefficients follows from the Gaussian upper bounds for Euler densities established by Bally and Talay (1996), since the preterminal time $1 - h \ge \frac{1}{2}$ is bounded away from zero.

Assumption 4.1 is compatible with the regular setting used to justify Euler density expansions in Bally and Talay (1995) and Bally and Talay (1996). It is not compatible with the square-root boundaries in the later financial application, where (A2) fails at the origin; that distinction is discussed in Section 8.

Theorem 4.2 (Local moment law)**.** *Under Assumption 4.1, for every fixed* $r > 1$*,*

$$h^{(r-1)K/2}\mathbb{E}[G_h^r] \longrightarrow c_{r,K}\, p(y \mid x)\, |V(y)|^{-(r-1)/2}, \tag{18}$$

where

$$c_{r,K} = (2\pi)^{-(r-1)K/2} r^{-K/2}. \tag{19}$$

In particular,

$$h^{K/2}\mathbb{E}[G_h^2] \longrightarrow (4\pi)^{-K/2} p(y \mid x) |V(y)|^{-1/2}. \tag{20}$$

Proof. Write $m_h(z) = y - z - h\mu(z)$, so that by (3)

$$g_h(y \mid z)^r = (2\pi h)^{-rK/2} |V(z)|^{-r/2} \exp\left[-\frac{r}{2h} m_h(z)^\intercal V(z)^{-1} m_h(z)\right], \tag{21}$$

and fix $\delta \in (0, \delta_0]$. Split

$$\begin{aligned} h^{(r-1)K/2}\mathbb{E}[G_h^r] &= I_h(\delta) + T_h(\delta), \\ I_h(\delta) &= h^{(r-1)K/2} \int_{B(y,\delta)} g_h(y \mid z)^r\, r_h(z \mid x)\, dz, \\ T_h(\delta) &= h^{(r-1)K/2} \int_{B(y,\delta)^c} g_h(y \mid z)^r\, r_h(z \mid x)\, dz. \end{aligned}$$

Tail term. For $z \notin B(y, \delta)$ and $h \le \delta/(4\bar{\mu} + 1)$, (A1) gives

$$\|m_h(z)\| \ge \|y - z\| - h\bar{\mu} \ge \tfrac{1}{2}\|y - z\| \qquad \text{and} \qquad \|m_h(z)\| \ge \tfrac{1}{2}\delta.$$

By (A2) we have $V(z)^{-1} \succeq \overline{v}^{-1} I_K$ and $|V(z)|^{-r/2} \le \underline{v}^{-rK/2}$, so

$$g_h(y \mid z)^r \le (2\pi h)^{-rK/2} \underline{v}^{-rK/2} \exp\left[-\frac{r\|m_h(z)\|^2}{2\overline{v}h}\right].$$

Splitting the exponent in half and using the two lower bounds separately,

$$g_h(y \mid z)^r \le (2\pi h)^{-rK/2} \underline{v}^{-rK/2} \exp\left[-\frac{r\delta^2}{16\overline{v}h}\right] \exp\left[-\frac{r\|y-z\|^2}{16\overline{v}h}\right].$$

Using (A4) and $\int_{\mathbb{R}^K} e^{-r\|w\|^2/(16\overline{v}h)} dw = (16\pi\overline{v}h/r)^{K/2}$, and collecting powers as $h^{(r-1)K/2} \cdot h^{-rK/2} \cdot h^{K/2} = 1$,

$$T_h(\delta) \le \bar{r}\,\underline{v}^{-rK/2}(2\pi)^{-rK/2}(16\pi\overline{v}/r)^{K/2} \exp\left[-\frac{r\delta^2}{16\overline{v}h}\right] \xrightarrow[h\downarrow 0]{} 0. \tag{22}$$

Local term. Substitute $z = z_h(u) = y + \sqrt{h}\,u$, so that $dz = h^{K/2}du$ and the domain becomes $\|u\| < \delta/\sqrt{h}$. Since $h^{(r-1)K/2}(2\pi h)^{-rK/2}h^{K/2} = (2\pi)^{-rK/2}$, every power of h cancels and

$$I_h(\delta) = (2\pi)^{-rK/2} \int_{\mathbb{R}^K} \mathbf{1}\{\|u\| < \delta/\sqrt{h}\}\, \Psi_h(u)\, du,$$

$$\Psi_h(u) = |V(z_h(u))|^{-r/2} r_h(z_h(u) \mid x) \exp\left[-\frac{r}{2h} m_h(z_h(u))^\top V(z_h(u))^{-1} m_h(z_h(u))\right].$$

Here $m_h(z_h(u)) = -\sqrt{h}\,u - h\mu(z_h(u))$, so

$$\frac{1}{h} m_h(z_h(u))^\top V(z_h(u))^{-1} m_h(z_h(u)) = \left(u + \sqrt{h}\,\mu(z_h(u))\right)^\top V(z_h(u))^{-1}\left(u + \sqrt{h}\,\mu(z_h(u))\right).$$

Fix u. Then $z_h(u) \to y$, so continuity of μ and V gives $V(z_h(u)) \to V(y)$ and $\sqrt{h}\,\mu(z_h(u)) \to 0$, and the quadratic form converges to $u^\top V(y)^{-1}u$. For h small enough that $\sqrt{h}\|u\| \le \delta_0$, (A5) and continuity of $p(\cdot \mid x)$ give $r_h(z_h(u) \mid x) \to p(y \mid x)$. Hence, pointwise in u,

$$\Psi_h(u) \longrightarrow |V(y)|^{-r/2} p(y \mid x) \exp\left[-\tfrac{r}{2} u^\top V(y)^{-1} u\right].$$

Domination. Let $h \le \min\{\delta_0, (2\bar{\mu})^{-2}\}$. If $\|u\| \ge 2\sqrt{h}\,\bar{\mu}$ then $\|u + \sqrt{h}\,\mu(z_h(u))\| \ge \frac{1}{2}\|u\|$, so by (A2) and (A4)

$$\Psi_h(u) \le \underline{v}^{-rK/2}\,\bar{r}\,\exp\left[-\frac{r\|u\|^2}{8\overline{v}}\right],$$

while if $\|u\| < 2\sqrt{h}\,\bar{\mu} \le 1$ then $\Psi_h(u) \le \underline{v}^{-rK/2}\bar{r}$. The dominating function

$$\Lambda(u) = \underline{v}^{-rK/2}\bar{r}\left\{\exp\left[-\frac{r\|u\|^2}{8\overline{v}}\right] + \mathbf{1}\{\|u\| \le 1\}\right\}$$

is integrable and does not depend on h. Dominated convergence therefore gives

$$\lim_{h\downarrow 0} I_h(\delta) = (2\pi)^{-rK/2}|V(y)|^{-r/2} p(y \mid x) \int_{\mathbb{R}^K} \exp\left[-\tfrac{r}{2} u^\top V(y)^{-1} u\right] du.$$

The Gaussian integral equals $(2\pi/r)^{K/2}|V(y)|^{1/2}$, so

$$\lim_{h\downarrow 0} I_h(\delta) = (2\pi)^{-rK/2}(2\pi/r)^{K/2} p(y \mid x)|V(y)|^{-(r-1)/2} = c_{r,K}\, p(y \mid x)|V(y)|^{-(r-1)/2}.$$

This limit does not depend on δ, and $T_h(\delta) \to 0$ by (22), which proves (18). Taking $r = 2$ and using $(2\pi)^{-K/2}2^{-K/2} = (4\pi)^{-K/2}$ gives (20). □

Remark 4.3 (Verification in exactly solvable cases)**.** The constant can be checked against closed forms. For standard Brownian motion $V \equiv I_K$, and substituting $p(y \mid x) = (2\pi)^{-K/2} e^{-\|y-x\|^2/2}$ into (18) reproduces the exact limit of Proposition 3.1 term by term. For a constant-coefficient diffusion $dY = \mu\, dt + \Sigma\, dW$ the same computation with $V = \Sigma\Sigma^{\top}$ reproduces (18) including the determinant factor. For a multidimensional Ornstein–Uhlenbeck process the preterminal law is Gaussian and the second moment is again available in closed form; Figure 4 confirms convergence to (20) numerically. The cases $r = 2$ and $r = 3$ supply exactly the second and third moments used in Theorem 4.7.

Remark 4.4 (Localisation)**.** Only three features of Assumption 4.1 are used: ellipticity and boundedness of μ near y, which drive the local term; the uniform bound (A4), which drives domination; and enough global control to make the tail negligible. Global uniform ellipticity is therefore stronger than necessary. It can be replaced by ellipticity on $B(y, \delta_0)$ together with any condition forcing the contribution of $\{Z_h \notin B(y, \delta)\}$ to vanish after multiplication by $h^{(r-1)K/2} \sup_z g_h(y \mid z)^r$, for instance a Gaussian upper bound of Aronson type on r_h. We state the global version because it is directly verifiable and because the content of the theorem is the dimensional constant, not the weakest hypotheses.

Remark 4.5 (Time-inhomogeneous coefficients)**.** If μ and V depend on time, the one-step kernel (3) evaluates them at time $1 - h$, and the proof goes through verbatim provided μ and V are jointly continuous at $(y, 1)$ and (A1) and (A2) hold uniformly in t on a neighbourhood of 1. The limit constant is unchanged and $V(y)$ is read as $V(y, 1)$. We suppress the time argument in the statement because carrying it obscures the cancellation of powers of h, which is where the dimensional constant comes from.

Corollary 4.6 (Generic variance scale)**.** *Let $q_M(y \mid x) = \mathbb{E}[G_h]$ and suppose $q_M(y \mid x) \to p(y \mid x)$. Then*

$$\operatorname{Var}(\widehat{q}_{M,S}) \sim \frac{\tau^2(x,y)}{Sh^{K/2}}, \qquad \tau^2(x,y) = (4\pi)^{-K/2} p(y \mid x) |V(y)|^{-1/2}. \tag{23}$$

Theorem 4.7 (Generic endpoint CLT)**.** *Let (M_n, S_n) satisfy $M_n \to \infty$ and $S_n \to \infty$, write $h_n = 1/M_n$, and suppose the effective endpoint size diverges:*

$$R_n = S_n h_n^{K/2} = \frac{S_n}{M_n^{K/2}} \longrightarrow \infty. \tag{24}$$

Then under Assumption 4.1,

$$\sqrt{R_n}\, \{\widehat{q}_{M_n,S_n}(y \mid x) - q_{M_n}(y \mid x)\} \Longrightarrow \mathcal{N}\Big(0, \tau^2(x,y)\Big), \tag{25}$$

where, as in (23)*,*

$$\tau^2(x,y) = (4\pi)^{-K/2} p(y \mid x) |V(y)|^{-1/2}. \tag{26}$$

This is a statement about the simulated transition density, centred at the Euler density q_{M_n} and not at the diffusion density p; Corollary 4.9 treats the latter. It is not a statement about any parameter estimator.

Proof. Define the triangular array

$$G_{n,s} = g_{h_n}(y \mid Z_{h_n,s}), \qquad s = 1, \dots, S_n,$$

where $Z_{h_n,1}, \dots, Z_{h_n,S_n}$ are the independent preterminal draws in (5). Within row n these are independent and identically distributed with common mean $\mathbb{E}G_{n,1} = q_{M_n}(y \mid x)$, and

$$\widehat{q}_{M_n,S_n} - q_{M_n} = \frac{1}{S_n} \sum_{s=1}^{S_n} (G_{n,s} - \mathbb{E}G_{n,s}).$$

Variance. Theorem 4.2 with $r = 2$ gives $h_n^{K/2}\mathbb{E}[G_{n,1}^2] \to \tau^2$. Since $\mathbb{E}G_{n,1} = q_{M_n} \to p(y \mid x)$ is bounded, $h_n^{K/2}(\mathbb{E}G_{n,1})^2 \to 0$, so

$$h_n^{K/2}\operatorname{Var}(G_{n,1}) \longrightarrow \tau^2, \qquad \operatorname{Var}(\widehat{q}_{M_n,S_n}) = \frac{\operatorname{Var}(G_{n,1})}{S_n} \sim \frac{\tau^2}{R_n}. \tag{27}$$

Third centred moment. Theorem 4.2 with $r = 3$ bounds the *raw* moment, $\mathbb{E}[G_{n,1}^3] = O(h_n^{-K})$. The centred moment does not follow from this automatically, and is obtained as follows. The summand is a Gaussian density value, hence nonnegative, so the elementary inequality $|a-b|^3 \le 4(a^3+b^3)$ for $a, b \ge 0$ applies and

$$\mathbb{E}|G_{n,1} - \mathbb{E}G_{n,1}|^3 \le 4\left\{\mathbb{E}[G_{n,1}^3] + (\mathbb{E}G_{n,1})^3\right\} = O(h_n^{-K}) + O(1) = O(h_n^{-K}), \tag{28}$$

using once more that q_{M_n} is bounded. Nonnegativity of the summand is what makes this step elementary; without it the centred third moment would require a separate argument.

Lyapunov condition. Write $\sigma_n^2 = \operatorname{Var}(G_{n,1})$, so $\sigma_n^2 \sim \tau^2 h_n^{-K/2}$ by (27). The Lyapunov ratio at order $2+\delta$ with $\delta = 1$ is

$$L_n = \frac{S_n\,\mathbb{E}|G_{n,1} - \mathbb{E}G_{n,1}|^3}{(S_n\sigma_n^2)^{3/2}} = \frac{S_n\,O(h_n^{-K})}{S_n^{3/2}\,\tau^3 h_n^{-3K/4}\{1+o(1)\}} = O\left(S_n^{-1/2}h_n^{-K/4}\right) = O\left(R_n^{-1/2}\right),$$

because $S_n^{-1/2}h_n^{-K/4} = (S_n h_n^{K/2})^{-1/2}$. Condition (24) gives $L_n \to 0$.

The Lyapunov central limit theorem for triangular arrays with row-wise independent summands therefore applies to $(S_n\sigma_n^2)^{-1/2}\sum_s(G_{n,s} - \mathbb{E}G_{n,s})$. Multiplying by $\sigma_n h_n^{K/4} \to \tau$ converts the normalisation $\sqrt{S_n}\,\sigma_n$ into $\sqrt{R_n}$ and yields (25). □

Remark 4.8 (Why a fixed-M central limit theorem cannot be reused)**.** For fixed M the summands do not change with S, their variance is a constant, and the classical $\sqrt{S}$ central limit theorem applies. The array above is genuinely triangular: the law of $G_{n,s}$ changes with n through h_n, and its variance diverges at rate $h_n^{-K/2}$. The correct normalisation is $\sqrt{R_n}$, which coincides with $\sqrt{S_n}$ only when $K = 0$. Substituting a fixed-M central limit theorem and then letting M grow, as in Lemma 3 of Brandt and Santa-Clara (2002), is exactly the step that fails; the limiting variance stated there is the variance of a summand that itself depends on M.

Corollary 4.9 (CLT centred at the diffusion density)**.** *Suppose in addition that*

$$q_M(y \mid x) - p(y \mid x) = b(x,y)h + o(h). \tag{29}$$

If

$$Sh^{K/2} \to \infty, \qquad \sqrt{Sh^{K/2}}\,h \to 0, \tag{30}$$

then

$$\sqrt{Sh^{K/2}}\,\{\widehat{q}_{M,S}(y \mid x) - p(y \mid x)\} \Longrightarrow \mathcal{N}(0, \tau^2(x,y)). \tag{31}$$

In terms of M and S, the second condition in (30) *is*

$$\frac{\sqrt{S}}{M^{1+K/4}} \longrightarrow 0. \tag{32}$$

For $K = 4$, the two conditions together define the feasible regime

$$M^2 \ll S \ll M^4. \tag{33}$$

Proof. Decompose

$$\sqrt{Sh^{K/2}}\,\{\widehat{q}_{M,S}-p\}=\sqrt{Sh^{K/2}}\,\{\widehat{q}_{M,S}-q_M\}+\sqrt{Sh^{K/2}}\,\{q_M-p\}\,.$$

The first term converges to $\mathcal{N}(0,\tau^2)$ by Theorem 4.7. By (29) the second term equals

$$\sqrt{Sh^{K/2}}\,\{b(x,y)h+o(h)\}\,,$$

which vanishes precisely under the second condition in (30). Slutsky's theorem gives (31).

For the restatement, $\sqrt{Sh^{K/2}}\,h=\sqrt{S}\,h^{K/4+1}=\sqrt{S}/M^{1+K/4}$, since $h=1/M$. This is (32), and squaring gives the equivalent form $S/M^{2+K/2}\to 0$.

For $K=4$ the first condition in (30) reads $S/M^2\to\infty$ and the second reads $S/M^4\to 0$, which is (33). The regime is nonempty: $S=M^3$ satisfies both. □

Remark 4.10 (Bias negligibility is not variance consistency)**.** The two conditions in (30) do different work and pull in opposite directions. The first, $Sh^{K/2}\to\infty$, makes the Monte Carlo error vanish and is what Theorem 4.7 needs; it is a lower bound on S relative to M. The second makes the Euler bias negligible *relative to the Monte Carlo standard deviation*, and because that standard deviation shrinks as S grows, it is an upper bound on S relative to M. Conflating them is the source of the difficulty in Table 1: the published condition $\sqrt{S}/M\to 0$ has the form of the second requirement but is imposed at the wrong power, and it then contradicts the first when $K\geq 4$.

Remark 4.11 (Dependency structure)**.** The density-level results depend on one another as follows. Assumption 4.1 supports Theorem 4.2, which is used only through its cases $r=2$ and $r=3$. The case $r=2$ gives the variance scale Corollary 4.6; the two cases together give the Lyapunov condition in Theorem 4.7. Corollary 4.9 adds the Euler bias expansion (29), which is an assumption imported from Bally and Talay (1995) and Bally and Talay (1996) rather than proved here, and combines it with Theorem 4.7. Proposition 5.1 uses Corollary 4.6 and (29) but no central limit theorem. In the Brownian benchmark, Proposition 3.1 replaces Theorem 4.2 with an exact computation and the bias expansion is unnecessary because the bias is zero, so Theorem 3.7 rests on no imported result at all. Theorem 3.3 is independent of every central limit theorem above; it uses only the exact Gaussian tail bound.

Remark 4.12 (The dimensional conflict)**.** The condition $\sqrt{S}/M\to 0$ suppresses an $O(M^{-1})$ Euler bias under an incorrectly fixed Monte Carlo scale. Once the variance inflation $M^{K/2}$ is included, the bias must be compared with the standard deviation $M^{K/4}/\sqrt{S}$, producing (32). In four dimensions, the lower consistency requirement and the upper bias requirement leave the nonempty interval (33). The published condition places S below the lower boundary of that interval.

5 Bias–variance trade-off and dimensional rate

Suppose the Euler density admits (29). Because (5) is unbiased for q_M,

$$\begin{aligned}\mathbb{E}\Big[(\widehat{q}_{M,S}-p)^2\Big]&=(q_M-p)^2+\mathrm{Var}(\widehat{q}_{M,S})\\&=b(x,y)^2h^2+\frac{\tau^2(x,y)}{Sh^{K/2}}+o\Big(h^2+\frac{1}{Sh^{K/2}}\Big)\,.\end{aligned}\tag{34}$$

Proposition 5.1 (Pointwise optimal discretisation)**.** *If $b(x,y)\neq 0$, the leading terms in* (34) *are balanced by*

$$h\asymp S^{-2/(K+4)},\qquad M\asymp S^{2/(K+4)}.\tag{35}$$

Table 1: Principal endpoint-density rate conditions.

Property	General dimension K	Four dimensions
Monte Carlo mean-square consistency	$S/M^{K/2} \to \infty$	$S/M^2 \to \infty$
Density CLT centred at q_M	$S/M^{K/2} \to \infty$	$S/M^2 \to \infty$
Euler bias negligible in corrected CLT	$\sqrt{S}/M^{1+K/4} \to 0$	$S/M^4 \to 0$
Condition in Brandt and Santa-Clara (2002)	$\sqrt{S}/M \to 0$	$S/M^2 \to 0$

The resulting optimal pointwise mean squared error is

$$\mathrm{MSE}_{\mathrm{opt}} \asymp S^{-4/(K+4)}. \tag{36}$$

Proof. Differentiate the leading expression $Bh^2 + C/(Sh^{K/2})$ with respect to h or equate its two terms. This gives $h^{K/2+2} \asymp S^{-1}$ and hence (35). Substitution yields (36). □

The rate is the familiar structure of a kernel-density problem: reducing the bandwidth controls approximation bias but decreases the effective number of simulations. The dimension enters through the volume of the final Gaussian kernel. Some illustrative exponents are

Dimension K	M_{opt}	Optimal MSE
1	$S^{2/5}$	$S^{-4/5}$
2	$S^{1/3}$	$S^{-2/3}$
4	$S^{1/4}$	$S^{-1/2}$
8	$S^{1/6}$	$S^{-1/3}$

This does not mean that M should always be chosen by pointwise MSE. Likelihood estimation requires accuracy over many endpoints and parameter values, and higher-order or bridge-based methods can change both constants and rates. It does mean that increasing M while holding S fixed, or while letting S grow too slowly relative to dimension, can make the simulation approximation worse rather than better.

6 From density error to log-likelihood error

For observations $Y_0, \ldots, Y_N$, the simulated log likelihood is a sum of terms

$$\log \widehat{q}_{M,S}(Y_{n+1} \mid Y_n; \theta).$$

Even when $\widehat{q}_{M,S}$ is unbiased for q_M, its logarithm is not unbiased. The effective endpoint size $R_{M,S} = Sh^{K/2}$ controls this nonlinearity.

Proposition 6.1 (Log-density delta expansion)**.** *Fix an endpoint with $q_M \to p > 0$ and suppose the conditions of Theorem 4.7 hold. If $R_{M,S} = Sh^{K/2} \to \infty$, then*

$$\sqrt{R_{M,S}}\,\{\log \widehat{q}_{M,S} - \log q_M\} \Longrightarrow \mathcal{N}\left(0, \frac{\tau^2}{p^2}\right), \tag{37}$$

and

$$\log \widehat{q}_{M,S} - \log q_M = \frac{\widehat{q}_{M,S} - q_M}{q_M} - \frac{(\widehat{q}_{M,S} - q_M)^2}{2q_M^2} + o_{\mathbb{P}}(R_{M,S}^{-1}). \tag{38}$$

Whenever the second-order expansion is uniformly integrable, it follows that

$$\mathbb{E}[\log \widehat{q}_{M,S}] - \log q_M = -\frac{\tau^2}{2p^2 R_{M,S}} + o(R_{M,S}^{-1}). \tag{39}$$

Proof. The first statement is the delta method applied to Theorem 4.7. Since $(\widehat{q} - q_M)/q_M = O_{\mathbb{P}}(R^{-1/2})$ and q_M is bounded away from zero, Taylor expansion of $\log(1+u)$ gives (38). Taking expectations requires uniform integrability of the remainder and lower-tail control; under that additional condition, the first-order term has zero expectation and the second-order term gives (39). □

The leading nonlinear scale is

$$R_{M,S}^{-1} = \frac{M^{K/2}}{S}, \tag{40}$$

not uniformly $1/S$. The statement in Brandt and Santa-Clara (2002, p. 171) that the logarithmic transformation induces a bias of order $1/S$ is valid with M fixed. It is not a joint-M, S order.

Suppose simulations are independent across observations and the local constants are uniformly bounded. For the average simulated criterion

$$\widehat{Q}_{N,M,S}(\theta) = \frac{1}{N}\sum_{n=0}^{N-1} \log \widehat{q}_{n,M,S}(\theta),$$

the stochastic average of the first-order simulation error has standard deviation of order

$$(NR_{M,S})^{-1/2},$$

while the average nonlinear bias remains of order $R_{M,S}^{-1}$. The simulation bias does not average out across observations.

A full estimator theorem requires uniform control in θ of densities, scores, and Hessians. The following standard perturbation statement separates those requirements from the endpoint calculations.

Proposition 6.2 (Abstract equivalence to exact MLE)**.** *Let $\widehat{\theta}_N$ maximise an exact average log likelihood Q_N, and let $\widehat{\theta}_{N,M,S}$ maximise a simulated criterion $\widehat{Q}_{N,M,S}$. Assume:*

(i) $\widehat{\theta}_N \to \theta_0$ and $\sqrt{N}(\widehat{\theta}_N - \theta_0)$ has the usual exact-MLE limit;

(ii) on a neighbourhood of θ_0,

$$\sup_\theta \|\nabla \widehat{Q}_{N,M,S}(\theta) - \nabla Q_N(\theta)\| = o_{\mathbb{P}}(N^{-1/2});$$

(iii) on the same neighbourhood,

$$\sup_\theta \|\nabla^2 \widehat{Q}_{N,M,S}(\theta) - \nabla^2 Q_N(\theta)\| = o_{\mathbb{P}}(1);$$

(iv) the exact criterion has a nonsingular limiting Hessian.

Then

$$\sqrt{N}(\widehat{\theta}_{N,M,S} - \widehat{\theta}_N) \to 0 \quad \textit{in probability.}$$

Proof. Expand the simulated first-order condition around $\widehat{\theta}_N$ and use the uniform score and Hessian bounds together with nonsingularity of the limiting exact Hessian. □

Under uniform analogues of the pointwise density expansion, plausible minimum requirements for first-order equivalence include

$$Sh^{K/2} \to \infty, \qquad \frac{\sqrt{N}}{Sh^{K/2}} \to 0, \qquad \sqrt{N}\, h \to 0, \tag{41}$$

corresponding respectively to vanishing score simulation noise, vanishing nonlinear simulation bias at the $N^{-1/2}$ scale, and negligible Euler bias. Derivatives of the simulated paths and Gaussian kernels can impose stronger conditions. Therefore (41) is a design implication, not a claimed complete theorem for arbitrary diffusions.

The broader Monte Carlo likelihood literature treats joint data and simulation limits through empirical-process, importance-sampling, or exact-simulation arguments rather than by combining two pointwise limits; see, among others, Lee (1995), Sung and Geyer (2007), Beskos, Papaspiliopoulos, and Roberts (2009), and Miasojedow et al. (2016).

7 What fails in the published asymptotic argument

This section distinguishes statements refuted by the Brownian example from steps that are merely unproved under the stated assumptions. Exactly one published statement is directly refuted, namely the joint transition-density convergence in Lemma 2. Everything else below is a gap in a proof, not a demonstrated falsehood.

7.1 Lemma 2: sequential convergence is used as joint convergence

The proof of Lemma 2 in Brandt and Santa-Clara (2002) applies the strong law to the Monte Carlo average and then invokes convergence of the Euler density. For every fixed M,

$$\widehat{q}_{M,S} \to q_M \quad \text{as } S \to \infty,$$

and separately

$$q_M \to p \quad \text{as } M \to \infty.$$

This gives the iterated limit

$$\lim_{M\to\infty} \lim_{S\to\infty} \widehat{q}_{M,S} = p.$$

It does not give convergence along every sequence $(M_n, S_n) \to (\infty, \infty)$. The summand changes with M and its variance diverges as $M^{K/2}$. Theorem 3.3 directly refutes the stated joint conclusion.

Two features of Lemma 2 are worth isolating. It is stated as a joint limit in M and S with no rate restriction whatsoever, so the refutation does not depend on the separate discussion of $\sqrt{S}/M \to 0$ below; and it is stated under Assumptions 1 and 2 alone, which standard Brownian motion satisfies. The counterexample therefore meets the lemma on its own terms.

Because Lemmas 3 to 6 take Lemma 2 as an input, and Theorem 1 is stated as the summary of Lemmas 1 to 6, the published proofs of Theorems 1 and 2 fail. That is a statement about the proofs. It does not establish that the conclusions of those theorems are false, and the subsections that follow identify further steps that are independently unproved.

7.2 Lemma 3: the fixed-M CLT is not uniform in M

The paper uses a $\sqrt{S}$ central limit theorem and requires $\sqrt{S}/M \to 0$ to suppress the Euler bias. The variance of the triangular-array summand is not asymptotically constant. It is of order $M^{K/2}$. The correct standardisation is $\sqrt{S/M^{K/2}}$. The Euler bias must then be compared with that scale, producing (32).

7.3 Lemma 4: an $O_{\mathbb{P}}$ term is treated as $o_{\mathbb{P}}$

Let $x_n = \widehat{q}_n - p_n$. A nondegenerate central limit theorem implies $x_n = O_{\mathbb{P}}(S^{-1/2})$ for fixed M, not $o_{\mathbb{P}}(S^{-1/2})$. The Taylor expansion

$$\log\left(1 + \frac{x_n}{p_n}\right) = \frac{x_n}{p_n} + O_{\mathbb{P}}(x_n^2)$$

retains a first-order stochastic term. In the joint limit, the correct order is $O_{\mathbb{P}}\{(Sh^{K/2})^{-1/2}\}$. Moreover, summing pointwise errors over N observations requires lower bounds or tail control for p_n, dependence control, and a triangular-array argument. These are not supplied.

7.4 Lemma 5: convergence of objective values does not control maximisers

An argmax result requires uniform convergence of the criterion and separation or local curvature around the maximiser. A bounded Hessian gives an upper quadratic bound. To infer that two maximisers are close from a small objective difference, one needs a lower quadratic bound or a negative Hessian whose smallest eigenvalue is bounded away from zero after normalisation. Pointwise convergence and identification alone are insufficient.

The structure of the argument makes this explicit. Summing the two second-order expansions leaves a difference of objective values on the left and a term proportional to $(\widehat{\theta}_N - \widehat{\theta}_{N,M,S})^2$ on the right, whose coefficient is an average of two Hessians. Assumption 4 of Brandt and Santa-Clara (2002) bounds that coefficient above. Dividing a small left-hand side by a coefficient that is only bounded above yields no bound at all on the squared distance; what is required is a bound below, that is, a uniformly negative definite Hessian in a neighbourhood. The conclusion drawn, $\widehat{\theta}_{N,M,S} - \widehat{\theta}_N = o(N^{1/2}/S^{1/4})$, also inherits whatever is wrong with the $o(N/S^{1/2})$ input from Lemma 4.

7.5 Lemmas 6 and 7: exact-MLE asymptotics require additional structure

Consistency of the exact diffusion MLE ordinarily requires a uniform law of large numbers, global or local identification, and stability assumptions such as stationarity and ergodicity when $N \to \infty$. A score expansion around θ_0 is not by itself a proof of consistency because the intermediate Hessian is evaluated between the estimator and θ_0.

Similarly, the martingale central limit theorem invoked from Hall and Heyde (1980) requires convergence of the realised conditional quadratic variation and a conditional Lindeberg condition. Defining Fisher information as an expectation does not make the realised conditional variance condition automatic. Boundedness of $\partial p/\partial\theta$ is also not a bound on the score $(\partial p/\partial\theta)/p$ when p can be small; strict positivity of p at each point is weaker than the uniform lower bound the argument needs.

These points do not imply that no correct exact-MLE theorem can be written. They imply that the short proof in Appendix A does not establish the stated result, even apart from the false transition-density joint limit.

7.6 What a counterexample to the argmax estimator would require

It is worth being explicit about the limit of the negative results above. Theorem 3.3 refutes a statement about the simulated transition density at one pair of points. It does not refute the conclusion of Theorem 1, which concerns the maximiser of the simulated log likelihood. Failure of a criterion pointwise does not imply failure of its argmax, and three obstacles separate the two.

First, the collapse is driven by a rare-event geometry that is not uniform over the state space or over the parameter. Theorem 3.3 fixes $x = y = 0$. An argmax argument needs control of $\widehat{q}_{M,S}(Y_{n+1} \mid Y_n; \theta)$ simultaneously over N observed pairs and over a neighbourhood of θ_0, and the sets on which individual densities collapse need not align.

Second, the simulated criterion is evaluated with common random numbers across θ, as the published implementation requires for a smooth objective. The summands are therefore strongly dependent across parameter values. A criterion that is badly behaved at each fixed θ may still have a well-behaved maximiser if the errors move together; conversely, common random numbers can create a random limit criterion whose argmax is random. Which happens is a property of the coupling, not of the pointwise limit.

Third, the logarithm is unbounded below. Collapse towards zero at a single observation drives one log-density term to $-\infty$, which may push the maximiser away from the region where collapse occurs rather than towards a wrong value. The direction of that effect depends on how the collapse region varies with θ.

A complete argmax counterexample would need to exhibit a parametric family, an admissible sequence (M_n, S_n), and either a limit criterion whose maximiser differs from θ_0, or a demonstration that the simulated criterion fails to converge uniformly on a neighbourhood in a way that the maximiser inherits. We record in Section B the approaches we attempted and where each fails. Until such a construction exists, the correct statement is that the published proofs of Theorems 1 and 2 do not establish their conclusions, not that those conclusions are false.

8 Mathematical status of the exchange-rate application

The empirical application estimates a four-dimensional system consisting of two interest rates, the log exchange rate, and exchange-rate volatility. It also reconstructs a market-incompleteness state. Several mathematical features are not covered by the generic SML theorem.

8.1 Finite-sample scaling diagnostics for the implementation

Before turning to the structure of the model it is worth recording how the implemented simulation design sits relative to the asymptotic ordering that the published theorems describe. Theorem 2 of Brandt and Santa-Clara (2002) imposes two conditions jointly,

$$\frac{\sqrt{S}}{M} \longrightarrow 0, \qquad \frac{N}{S^{1/4}} \longrightarrow 0, \tag{42}$$

the second entering through the $o(N^{1/2}/S^{1/4})$ bound of their Lemma 5. A sequence satisfying both must eventually have $N^4 \ll S \ll M^2$.

What follows is a set of diagnostics, not a test. A single finite triple (N, M, S) cannot satisfy or violate a limit; only a sequence can. The question a diagnostic can answer is narrower and still useful: are the finite ratios appearing in (42) small, so that the implementation resembles a point along such a sequence, and does the theorem supply any error bound at the implemented sizes? The answer to the second question is no in any case, because (42) is a statement about limits and carries no quantitative finite-sample content.

The estimation uses $N = 544$ weekly observations from January 1990 to May 2000, $M = 10$ Euler substeps, and $S = 5{,}000$ simulations together with 5,000 antithetic variates. Table 2 reports the ratios. At the implemented values $\sqrt{S}/M = 7.07$ and $N/S^{1/4} = 64.7$. Neither is small, so the implementation does not resemble a design drawn from a sequence along which

Table 2: Finite-sample scaling ratios for the implemented design and three variations, at $N = 544$ and $K = 4$. The effective endpoint size is $R_{M,S} = S/M^2$ from Definition 2.1. These are diagnostics: the columns record how the implementation is placed, not whether any limit holds.

Scenario	M	S	$R_{M,S} = S/M^2$	$\sqrt{S}/M$	$N/S^{1/4}$
Implemented	10	5,000	50.0	7.07	64.7
Double M and S	20	10,000	25.0	5.00	54.4
Double M, S fixed	20	5,000	12.5	3.54	64.7
Double M, quadruple S	20	20,000	50.0	7.07	45.7

(42) is operating. That is the whole of the claim; nothing here shows that the estimates are inaccurate.

The antithetic variates deserve a separate word. They are a variance-reduction device: each antithetic draw is generated from $-\varepsilon$ where its partner uses ε, so the two are perfectly negatively dependent and the pair contributes less than two independent draws to any variance calculation. Counting them as though they doubled S would be wrong in both directions, understating the variance reduction they achieve for smooth integrands and overstating the number of independent endpoint draws. Table 2 therefore counts $S = 5{,}000$.

The dimensional diagnostic of Definition 2.1 gives the complementary reading. With $K = 4$, $M = 10$ and $S = 5{,}000$,

$$R_{M,S} = \frac{S}{M^{K/2}} = \frac{5{,}000}{100} = 50. \tag{43}$$

Each transition density in a 544-term log likelihood is therefore resolved by an effective endpoint sample of order 10^2, not 10^4. Again this is a scaling statement and not a verdict: an effective size of 50 may well be adequate at these parameter values, and the exact Brownian formulas of Section 3 would be the way to check that for a given target density.

The last three rows of Table 2 record how $R_{M,S}$ responds to refinement, because the arithmetic is easy to get wrong. In four dimensions $R_{M,S} = S/M^2$, so

$$R_{2M,2S} = \frac{2S}{4M^2} = \frac{R_{M,S}}{2}, \qquad R_{2M,S} = \frac{R_{M,S}}{4}, \qquad R_{2M,4S} = R_{M,S}. \tag{44}$$

The paper reports a final round of optimisation in which both S and M are doubled. That halves the effective endpoint size rather than increasing it, and the reported stability of the estimates under this refinement is therefore not evidence that the simulation is converging in the endpoint dimension; it is a comparison between two designs, the second of which has half the effective size of the first. Holding $R_{M,S}$ fixed while doubling M requires quadrupling S, which is the last row of the table.

8.2 Square-root coefficients violate the generic assumptions

The interest rates are modelled as correlated Cox–Ingersoll–Ross processes (Cox et al., 1985),

$$dr_t = \lambda(\vartheta - r_t)dt + \sigma\sqrt{r_t}\,dW_t, \qquad dr_t^* = \lambda^*(\vartheta^* - r_t^*)dt + \sigma^*\sqrt{r_t^*}\,dW_t^*. \tag{45}$$

The generic theory assumes infinitely differentiable coefficients with bounded derivatives and a positive-definite covariance matrix. The function $r \mapsto \sqrt{r}$ is not differentiable at zero, has unbounded derivatives near zero, is not defined on the negative half-line as a real coefficient, and makes the covariance degenerate at the boundary. The same singularity appears explicitly

Table 3: Feller ratios at the estimates reported in Table 3 of Brandt and Santa-Clara (2002). For a process $dZ = \kappa(\vartheta_Z - Z)dt + \sigma_Z\sqrt{Z}\,dB$ the ratio is $2\kappa\vartheta_Z/\sigma_Z^2$, and the origin is unattainable if and only if it is at least one. For the incompleteness state the ratio equals 1/2 for all admissible parameters.

System	State	κ	ϑ_Z	σ_Z	$2\kappa\vartheta_Z/\sigma_Z^2$
U.S. vs. U.K.	r (U.S.)	0.284	0.053	0.028	38.4
	r^* (U.K.)	0.486	0.074	0.056	22.9
	ϵ^2	0.640	0.0121	0.176	0.5
U.S. vs. Germany	r (U.S.)	0.305	0.058	0.027	48.5
	r^* (Germany)	0.088	0.064	0.042	6.4
	ϵ^2	0.676	0.0151	0.202	0.5

in the transformed system of Appendix B of Brandt and Santa-Clara (2002), whose drift for $\sqrt{r_t}$ contains the term $-\frac{1}{8}\sigma^2 r_t^{-1/2}$.

It is important to separate this structural violation from its practical consequences, and the two point in opposite directions for the interest rates. Table 3 reports the Feller ratio $2\lambda\vartheta/\sigma^2$ at the estimates of Table 3 of Brandt and Santa-Clara (2002). It ranges from 6.4 to 48.5, so in every case the ratio comfortably exceeds one, the origin is unattainable for the exact processes, and the interest-rate states are far from the boundary. The corresponding Euler negativity probabilities are numerically nil: with $h = 1/520$, corresponding to weekly data and $M = 10$, a step taken from the German long-run mean $\vartheta^* = 0.064$ has mean-to-standard-deviation ratio 137, and even a step taken from the implausibly low state $r^* = 0.001$ has ratio 17.4. The assumptions of the generic theorem are genuinely violated by (45), and the estimator applied is therefore not the estimator analysed; but for r and r^* that gap is a matter of proof coverage rather than of numerical damage.

The market-incompleteness state behaves differently. Let $X_t = \epsilon_t^2$. Equation (37) of Brandt and Santa-Clara (2002) specifies

$$dX_t = (\beta^2 - 2\alpha X_t)dt + 2\beta\sqrt{X_t}\,dY_t, \tag{46}$$

which is a CIR process with $\kappa = 2\alpha$, long-run mean $\beta^2/(2\alpha)$, and diffusion coefficient 2β. Its Feller ratio is

$$\frac{2\kappa\vartheta_X}{\sigma_X^2} = \frac{2\beta^2}{4\beta^2} = \frac{1}{2}$$

for every $\alpha > 0$ and every $\beta \neq 0$. The ratio is exactly one half independently of the parameter values, so no estimate can move this process away from its boundary: zero is attainable, and the instantaneous covariance degenerates there. This is a structural feature of the specification, not a property of a particular estimate. Proposition 8.2 below shows that the boundary is not merely attainable in principle but attained exactly in the reported results.

8.3 The simulated state and the reality of the incompleteness process

The previous subsection treats (46) as a state in its own right, which is how the model presents it. The estimator does not. Equation (31) and Appendix B of Brandt and Santa-Clara (2002) eliminate ϵ_t using the variance identity and simulate the four-dimensional state

$$(r_t,\ r_t^*,\ e_t,\ v_t), \tag{47}$$

where e_t is the log exchange rate and v_t is its instantaneous volatility, proxied by a one-week at-the-money implied volatility. The incompleteness state is therefore never discretised directly.

This relocates the boundary problem rather than removing it, and the relocated version is the one that binds.

The identity that eliminates ϵ_t is

$$\epsilon_t = \sqrt{v_t^2 - (\phi_t^2 + \phi_t^{*2} - 2\rho_{ww^*}\phi_t\phi_t^*) - (\psi_t^2 + \psi_t^{*2} - 2\rho\,\psi_t\psi_t^*)}, \tag{48}$$

where $\phi_t = \phi\sqrt{r_t}$ and $\phi_t^* = \phi^*\sqrt{r_t^*}$ are the market prices of domestic and foreign interest-rate risk and ψ_t, ψ_t^* are the market prices of pure currency risk. The state ϵ_t appears in the diffusion of the log exchange rate, through $v_t\,dX_t = \phi_t dW_t - \phi_t^* dW_t^* + \psi_t dZ_t - \psi_t^* dZ_t^* + \epsilon_t dU_t$, and in the coefficients of the v_t equation obtained from Itô's lemma. Every Euler substep of (47) therefore requires the radicand of (48) to be nonnegative at the simulated state, not merely at the observed data. Ordinary Euler discretisation of a system whose coefficients are defined by (48) offers no such guarantee: the increments of v_t are conditionally Gaussian and unbounded below, while the subtracted terms are continuous and generically bounded away from zero.

This is the operative form of the degeneracy, and three features make it sharp rather than hypothetical. First, the constraint is not slack: Proposition 8.2 shows that the reported identification scheme drives $\min_t \epsilon_t^2$ to exactly zero, and Table 4 of Brandt and Santa-Clara (2002) reports a minimum excess volatility of 0.0000 in all six reported specifications. The simulated system is started, at least once in every sample, from a state exactly on the boundary. Second, the failure is not a recoverable numerical event such as a negative variance that can be floored at zero; it makes the drift and diffusion coefficients of the next substep complex-valued, so the chain is undefined rather than inaccurate. Third, the paper reports assigning zero likelihood to parameter values that violate $\epsilon_t^2 \geq 0$ at the observed dates, which is a restriction on parameters, not a rule for what the simulator does when an intermediate Euler state violates it.

As with the interest rates, a truncation, reflection, absorption, or exact-transition rule would resolve the issue, but each such rule defines a different approximating chain with a different transition density and requires its own convergence analysis. The published paper does not specify one. Figure 5 illustrates the mechanism on the incompleteness specification (46) taken at face value, where it can be computed in closed form: for an Euler step of size h the probability of a negative next state from $X_t = x > 0$ is

$$\Phi\left(-\frac{x + (\beta^2 - 2\alpha x)h}{2\beta\sqrt{xh}}\right). \tag{49}$$

The figure is an illustration of the class of failure, not a simulation of the implemented chain, whose boundary is (48) and whose exact crossing probability depends on the drift and diffusion of v_t derived in Appendix B.

8.4 The time-varying volatility specification is implicit

The time-varying pure-currency risk prices have the form

$$\psi_t = a_t + bv_t, \qquad \psi_t^* = a_t^* + b^*v_t, \tag{50}$$

where a_t and a_t^* depend on the interest-rate differential and log exchange rate. The variance identity is

$$v_t^2 = D_t + \psi_t^2 + (\psi_t^*)^2 - 2\rho\psi_t\psi_t^*, \tag{51}$$

where D_t collects the interest-rate risk term and ϵ_t^2.

Substituting (50) into (51) gives

$$(1 - \kappa_b)v_t^2 - 2\eta_t v_t - \zeta_t = 0, \tag{52}$$

where

$$\kappa_b = b^2 + (b^*)^2 - 2\rho b b^*, \tag{53}$$

$$\eta_t = a_t b + a_t^* b^* - \rho(a_t b^* + a_t^* b), \tag{54}$$

$$\zeta_t = D_t + a_t^2 + (a_t^*)^2 - 2\rho a_t a_t^*. \tag{55}$$

When the underlying two-dimensional correlation matrix is positive semidefinite and $D_t \geq 0$, one has $\zeta_t \geq 0$.

Proposition 8.1 (Branch structure of the implicit volatility)**.** *Let $\zeta_t > 0$.*

(i) If $\kappa_b < 1$, equation (52) has exactly one positive root and one negative root, so the positive branch is unique.

(ii) If $\kappa_b > 1$, equation (52) has either no positive root or exactly two positive roots. A unique positive root is impossible except in the knife-edge case $\eta_t^2 = (\kappa_b - 1)\zeta_t$ of a repeated root. Specifically, positive roots exist if and only if $\eta_t < 0$ and $\eta_t^2 \geq (\kappa_b - 1)\zeta_t$, and they are then

$$v_t^{\pm} = \frac{-\eta_t \pm \sqrt{\eta_t^2 - (\kappa_b - 1)\zeta_t}}{\kappa_b - 1}.$$

Proof. Rewrite (52) as $(\kappa_b - 1)v_t^2 + 2\eta_t v_t + \zeta_t = 0$.

(i) If $1 - \kappa_b > 0$ the product of the roots of (52) is $-\zeta_t/(1 - \kappa_b) < 0$, so the real roots have opposite signs. The discriminant is $4\{\eta_t^2 + (1 - \kappa_b)\zeta_t\} > 0$, hence exactly one root is positive.

(ii) If $\kappa_b - 1 > 0$ the product of the roots is $\zeta_t/(\kappa_b - 1) > 0$ and their sum is $-2\eta_t/(\kappa_b - 1)$. A positive product excludes roots of opposite signs and excludes a zero root, so the real roots, when they exist, are either both positive or both negative, and they are both positive precisely when the sum is positive, that is when $\eta_t < 0$. Real roots exist when the discriminant $4\{\eta_t^2 - (\kappa_b - 1)\zeta_t\}$ is nonnegative, and the stated formula is the quadratic formula. Equality of the two roots requires $\eta_t^2 = (\kappa_b - 1)\zeta_t$, a codimension-one condition on the state. □

Table 4 of Brandt and Santa-Clara (2002) reports, for model C,

$$(b, b^*, \rho) = (1.514, 2.581, 0.89)$$

for the U.S.–U.K. system. These values imply

$$\kappa_b \approx 1.9982, \qquad 1 - \kappa_b \approx -0.9982.$$

These estimates fall in case (ii) of Proposition 8.1, not merely outside case (i). For the U.S.–U.K. model-C specification the volatility identity therefore admits either no positive solution or two positive solutions at almost every state, and a unique positive volatility branch is not available. Which of the two branches is intended is not determined by the specification, and the branch structure is state dependent: as η_t crosses zero, or as $\eta_t^2 - (\kappa_b - 1)\zeta_t$ changes sign, the solution set changes between empty and a pair. Since $\zeta_t \geq 0$ requires only positive semidefiniteness of the two-dimensional correlation matrix and $D_t \geq 0$, both of which hold by construction, this is not a knife-edge consequence of a particular parameter estimate; it follows from $\kappa_b > 1$ alone. For the U.S.–Germany estimates $\kappa_b \approx 0.0702 < 1$ and case (i) applies, so that system does have a unique positive branch. The calculation is shown in Table 4.

This observation does not prevent the authors from evaluating (51) conditional on an observed implied volatility, which is what the estimation does at the sample dates. It matters when the equations are interpreted as a generative four-dimensional diffusion whose state v_t must exist uniquely and evolve continuously, which is what the simulation of (47) requires at every Euler substep. Appendix B defines the drift and diffusion of v_t only implicitly, with those coefficients appearing inside the formulas used to define them. An invertibility and branch selection condition is required.

Table 4: Quadratic coefficient implied by the model-C volatility loadings reported in Table 4 of Brandt and Santa-Clara (2002). Here $\kappa_b = b^2 + (b^*)^2 - 2\rho bb^*$.

Market	b	b^*	ρ	κ_b	$1-\kappa_b$
U.S. vs. U.K.	1.514	2.581	0.89	1.9982	−0.9982
U.S. vs. Germany	−0.142	0.127	0.94	0.0702	0.9298

8.5 Pairwise correlations do not guarantee a valid Brownian system

The application specifies an instantaneous correlation matrix for the four Brownian motions (W_t, W_t^*, X_t, Y_t). Write it in block form as

$$R_t = \begin{pmatrix} A_t & r_t \\ r_t^\top & 1 \end{pmatrix}, \tag{56}$$

where A_t is the 3×3 block for (W_t, W_t^*, X_t) and r_t contains the three correlations with Y_t. A valid Brownian system requires $R_t \succeq 0$ for every state and parameter value. If $A_t \succ 0$, the Schur-complement condition is

$$1 - r_t^\top A_t^{-1} r_t \geq 0. \tag{57}$$

Constraining each entry of r_t to lie in $[-1, 1]$ is not sufficient. The paper does not state a Cholesky, partial-correlation, hyperspherical, or Schur-complement parameterisation that enforces (57). It is possible that the numerical implementation rejected invalid matrices, but such a rule is not part of the mathematical specification.

8.6 Minimum incompleteness is an identifying selection rule

The exchange-rate dynamics do not separately identify one risk-premium component and the degree of market incompleteness. The paper resolves this by choosing the unidentified quantity, described as chosen “in a least-squares sense,” to minimise $\sum_t \epsilon_t^2$ subject to $\epsilon_t^2 \geq 0$ at every date. Under constant pure-currency risk prices the free quantity is the product $\sigma_{z^*}\epsilon_{c^*}$; under time-varying prices it is the correlation ρ_{zz^*}.

The description as a least-squares problem is misleading in a way that determines the answer. Because ϵ_t^2 is itself the residual, the objective $\sum_t \epsilon_t^2$ is linear, not quadratic, in the free quantity, so the solution is always a corner of the feasible set rather than an interior least-squares projection.

Proposition 8.2 (Minimum-incompleteness solution)**.** *Let A_t denote the observed exchange-rate variance after subtracting all already identified components and let the free quantity enter linearly, so that $\epsilon_t^2 = A_t - \gamma_t c$ for known loadings γ_t and a scalar c.*

(i) Constant risk prices. Here $\gamma_t \equiv 1$ and $c = \sigma_z\epsilon_c + \sigma_{z^*}\epsilon_{c^*}$. The problem

$$\min_c \sum_{t=1}^{N} (A_t - c) \quad \textit{subject to} \quad A_t - c \geq 0 \quad (t = 1, \ldots, N) \tag{58}$$

has the unique solution

$$c^* = \min_{1 \leq t \leq N} A_t, \qquad \epsilon_t^2 = A_t - \min_s A_s. \tag{59}$$

(ii) Time-varying risk prices. Here $c = \rho_{zz^*}$ and $\gamma_t = -2\psi_t\psi_t^*$, so the objective is again linear in c with slope $-\sum_t \gamma_t$. Provided $\sum_t \gamma_t \neq 0$, the optimum is again attained at a corner of the feasible polyhedron $\{c : A_t - \gamma_t c \geq 0 \ \forall t\}$, at which at least one constraint is active.

In both cases $\min_t \epsilon_t^2 = 0$ *exactly, unless the corner is attained only in a limiting or numerically constrained sense.*

Proof. (i) The objective equals $\sum_t A_t - Nc$ and is strictly decreasing in c. Feasibility requires $c \leq \min_t A_t$. Hence the largest feasible value is the unique optimum, and the constraint indexed by $\arg\min_t A_t$ is active.

(ii) The objective equals $\sum_t A_t - c\sum_t \gamma_t$, which is affine in c with nonzero slope, and the feasible set is a closed interval determined by the constraints $A_t - \gamma_t c \geq 0$. An affine function with nonzero slope on a closed interval attains its minimum at an endpoint, at which the corresponding constraint holds with equality. □

The prediction is verifiable in the published output. Table 4 of Brandt and Santa-Clara (2002) reports the minimum of the inferred excess volatility as 0.0000 in every one of the six reported specifications, that is for models A, B, and C in both the U.S.–U.K. and the U.S.–Germany systems. Part (ii) of the proposition is what accounts for models B and C, where the free quantity is a correlation rather than a level, and where the paper separately reports that the constraint is binding.

The mechanism is acknowledged in part in the original paper, which observes on p. 193 that with constant risk prices the inferred excess volatility is essentially $\epsilon_t = \{v_t^2 - \min_s v_s^2\}^{1/2}$, and draws from this the conclusion that a volatile v_t leaves a large average residual. Proposition 8.2 formalises that observation, shows that it is a property of the optimisation rather than of the data, and extends it to the time-varying specifications where the same corner structure holds for a different free parameter. The consequence for interpretation is that the volatility of volatility, rather than its level, mechanically drives much of the estimated excess volatility: the residual is the observed variance minus its sample minimum after other adjustments, so its average size is a measure of the dispersion of v_t^2 and is bounded below by construction.

The rule is transparent, but it is not identification by the likelihood of the original structural model. It selects a member of an observationally equivalent or weakly identified class by imposing the smallest feasible incompleteness. The paper describes it as a "prior of sorts." Mathematically, the final estimator combines simulated likelihood, auxiliary least squares from bond yields, a minimum-incompleteness corner rule, and inequality constraints. A further consequence, used in Section 8.3, is that the reconstructed incompleteness state sits exactly on its own degenerate boundary at the minimising date.

8.7 Binding constraints imply nonregular inference

The empirical optimisation assigns zero likelihood to parameter values for which any reconstructed ϵ_t^2 is negative. The paper later states that the constraint is binding in the time-varying specifications. A criterion with an active inequality boundary is not twice continuously differentiable across the boundary, and the usual unconstrained normal approximation need not hold. Boundary MLEs and likelihood-ratio statistics commonly have tangent-cone or mixture limits rather than the standard interior distributions; see Self and Liang (1987).

Stacking first-order conditions and reporting ordinary GMM standard errors does not automatically account for:

- a sample-dependent feasible set defined by N inequalities;
- changes in the active set;
- nondifferentiability of an inner constrained optimisation;
- structural parameters selected through minimum incompleteness;
- an optimum on or near the boundary.

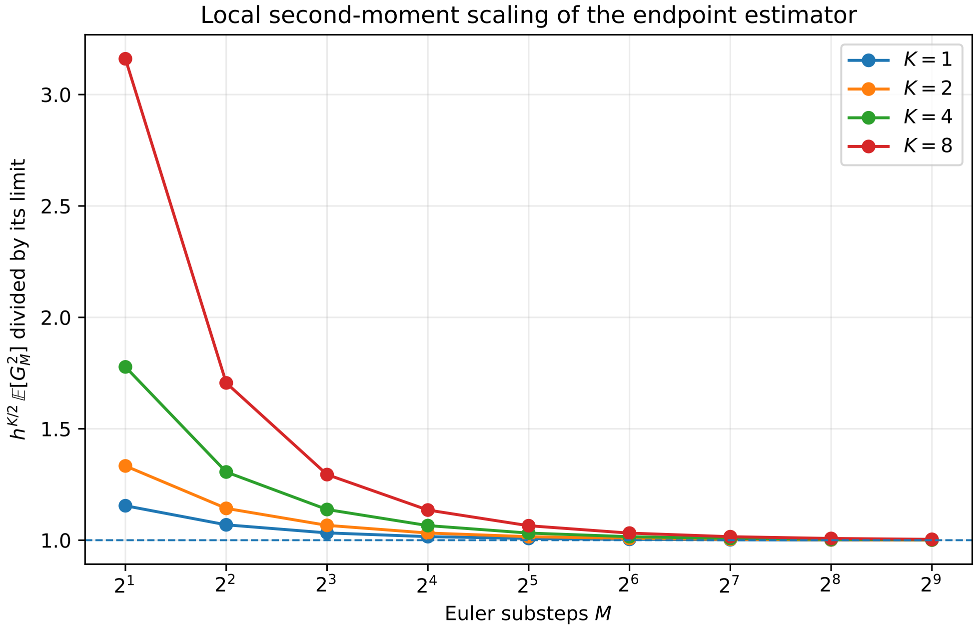

Figure 1: Exact local second-moment scaling for standard Brownian motion. The plotted quantity converges to one after division by its theoretical limit.

A separate constrained-estimation analysis is needed before conventional symmetric standard errors and Wald tests can be justified.

9 Numerical evidence

All numerical results in this section are generated by the accompanying Python script. They use exact Gaussian benchmarks where possible and are intended to illustrate the theorems rather than replace them.

9.1 Second-moment scaling

Figure 1 plots

$$h^{K/2}\mathbb{E}[G_h^2]$$

divided by its limiting value for $K \in \{1, 2, 4, 8\}$. The exact formula (11) converges to the constant predicted by Theorem 4.2. The unscaled second moment grows as $M^{K/2}$.

9.2 Three joint asymptotic paths

For $K = 4$, Figure 2 compares the exact relative root mean squared error under $S = M$, $S = M^2$, and $S = M^3$. The first path satisfies the published condition but has increasing RMSE. The second keeps the effective endpoint size constant and does not become consistent. The third satisfies $S/M^2 \to \infty$ and has decreasing RMSE.

9.3 Typical estimates collapse while the mean remains correct

Figure 3 simulates the counterexample path $M = S$. The sample median falls rapidly towards zero, and by $M = 512$ more than 95% of estimates are below half the true density. The sample

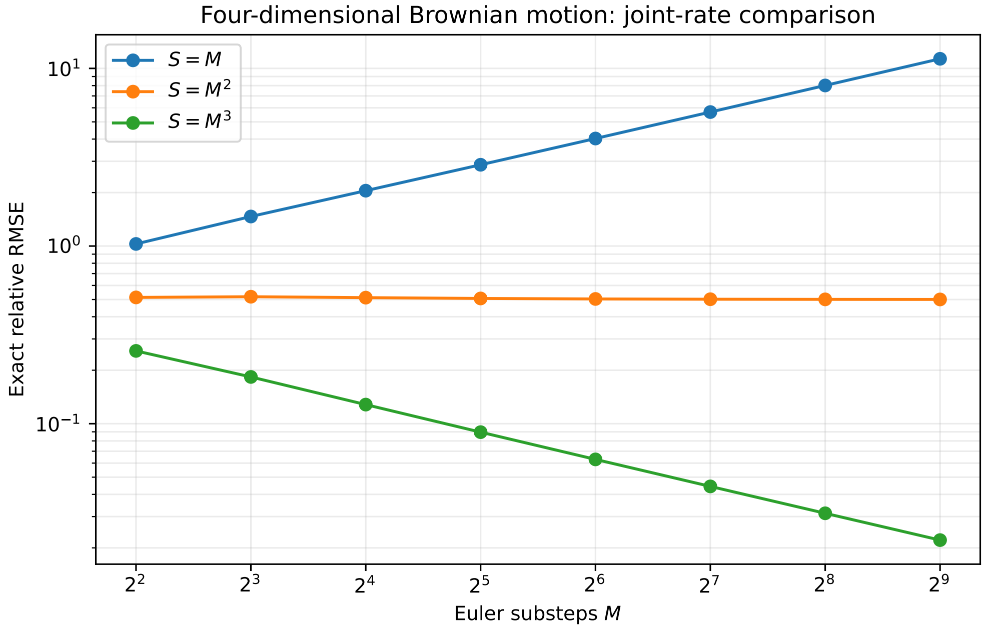

Figure 2: Exact relative RMSE at the origin for four-dimensional Brownian motion.

Table 5: Simulation of the four-dimensional Brownian counterexample path $M = S$. The estimator is divided by the true density.

M	Replications	Mean	Median	10th pct.	90th pct.	$\mathbb{P}(\widehat{q} < 0.5p)$
8	20000	1.001	0.3674	1.427×10^{-2}	2.940	0.560
16	20000	1.009	0.1582	1.520×10^{-3}	3.161	0.665
32	20000	1.027	0.03628	5.925×10^{-5}	2.746	0.754
64	20000	1.002	0.004223	8.507×10^{-7}	1.777	0.827
128	20000	1.042	1.537×10^{-4}	1.487×10^{-9}	0.7409	0.886
256	19531	0.941	1.118×10^{-6}	7.168×10^{-14}	0.1579	0.929
512	9765	0.998	8.604×10^{-10}	7.076×10^{-20}	0.01738	0.955

mean remains near one after normalisation because a small number of very large realisations carry the expectation.

9.4 A nonzero drift does not remove the dimensional law

Consider the diagonal Ornstein–Uhlenbeck process

$$dY_t = -Y_t dt + dW_t$$

in four dimensions. Its exact transition density is known, and the preterminal Euler state is Gaussian, so the second moment of the endpoint estimator can again be computed exactly. Figure 4 shows convergence to the same local constant in Theorem 4.2. The phenomenon is therefore not specific to driftless Brownian motion.

9.5 Euler boundary violations in the incompleteness process

Figure 5 evaluates (49) using the reported estimates $(\alpha, \beta) = (0.320, 0.088)$ and $(0.338, 0.101)$. Even with relatively fine steps, an ordinary Euler chain near the attainable zero boundary can

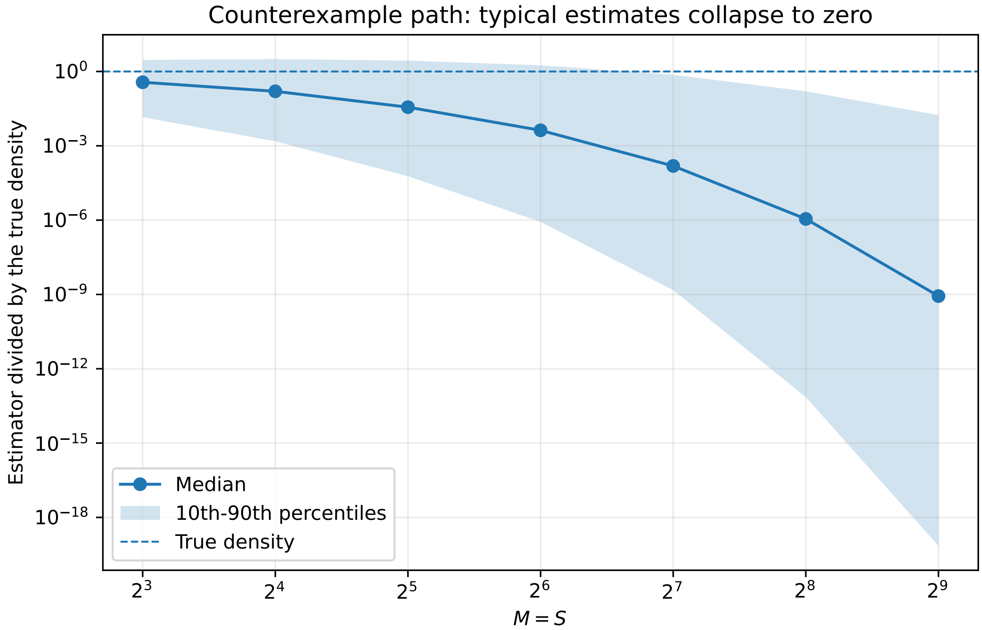

Figure 3: Monte Carlo distribution along $M = S$ for four-dimensional Brownian motion. Values are divided by the true density. The dashed line is one.

become negative with non-negligible probability. Once it does, the next square-root coefficient is not real without an additional numerical convention.

As explained in Section 8.3, this calculation applies to the incompleteness specification (46) treated as a state in its own right. The implemented chain simulates (r, r^*, e, v) and recovers ϵ_t from (48), so the boundary that must not be crossed is the complete-markets surface rather than $\{X = 0\}$. The figure is therefore an exact illustration of the mechanism rather than of the implemented chain. The contrast with the interest rates in Table 3 is the substantive content: for r and r^* the Feller ratios exceed one by an order of magnitude and the boundary is never approached, whereas for the incompleteness state the ratio is exactly $1/2$ for all parameter values and, by Proposition 8.2, the boundary is reached exactly.

10 Remedies and alternative constructions

The endpoint variance problem is not an unavoidable property of likelihood inference for diffusions. It is a consequence of drawing preterminal states from the unconditioned forward Euler law and asking a shrinking final Gaussian kernel to connect them to a fixed observed endpoint.

10.1 Bridge and importance proposals

A bridge proposal guides simulated paths towards the observed endpoint. In importance-sampling language, it increases the probability of drawing paths in the region that contributes to the transition-density integral and corrects the proposal through a likelihood ratio. This is the direction taken by the acceleration methods studied by Durham and Gallant (2002), the path-augmentation schemes of Elerian et al. (2001), conditioned-diffusion methods such as Delyon and Hu (2006), the regularised proposal of Lindström (2012), and later guided proposals for multidimensional bridges (Schauer et al., 2017; Meulen and Schauer, 2017). The

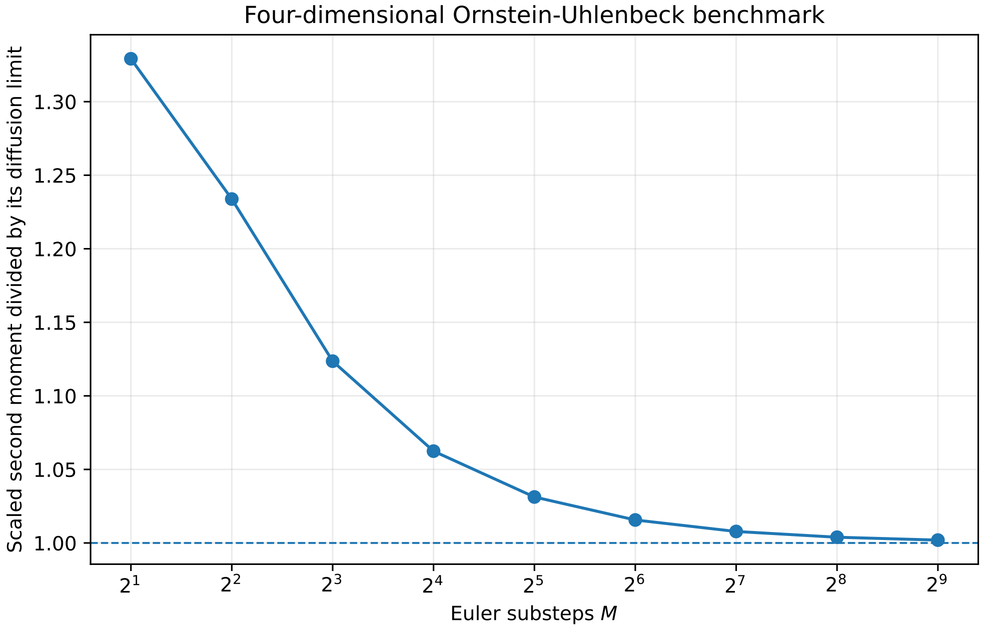

Figure 4: Scaled second moment for a four-dimensional Ornstein–Uhlenbeck diffusion.

nonparametric simulation approach of Nicolau (2002) replaces the endpoint Gaussian kernel by a density estimate, which changes the bandwidth problem rather than removing it.

A useful proposal should be assessed through the moments of its importance weights. Merely changing the path simulator does not guarantee finite or dimension-stable variance, but conditioning on the endpoint directly targets the rare-event failure identified here. The numerical comparisons of Stramer and Yan (2007) are consistent with this reading: the endpoint estimator degrades as M grows at fixed S, which is the finite-sample signature of Corollary 4.6.

10.2 Exact or discretisation-free likelihood estimators

For restricted classes of diffusions, exact simulation and unbiased transition-density estimators remove Euler bias entirely. Beskos, Papaspiliopoulos, Roberts, and Fearnhead (2006) and Beskos, Papaspiliopoulos, and Roberts (2009) develop likelihood-based methods of this kind. Their applicability is model dependent, but they illustrate the correct theoretical structure: unbiasedness, continuity in parameters, finite-moment conditions, and joint control of data and Monte Carlo sample sizes are stated explicitly.

10.3 Analytical density expansions

Closed-form expansions, such as those of Aït-Sahalia (2002), avoid endpoint Monte Carlo variance but introduce an approximation-order problem of their own. They can be attractive for models that can be transformed to satisfy the required regularity conditions.

10.4 Positivity-preserving discretisation

For square-root components, ordinary Euler should be replaced by an exact transition where available or by a specified positivity-preserving approximation. The choice is part of the

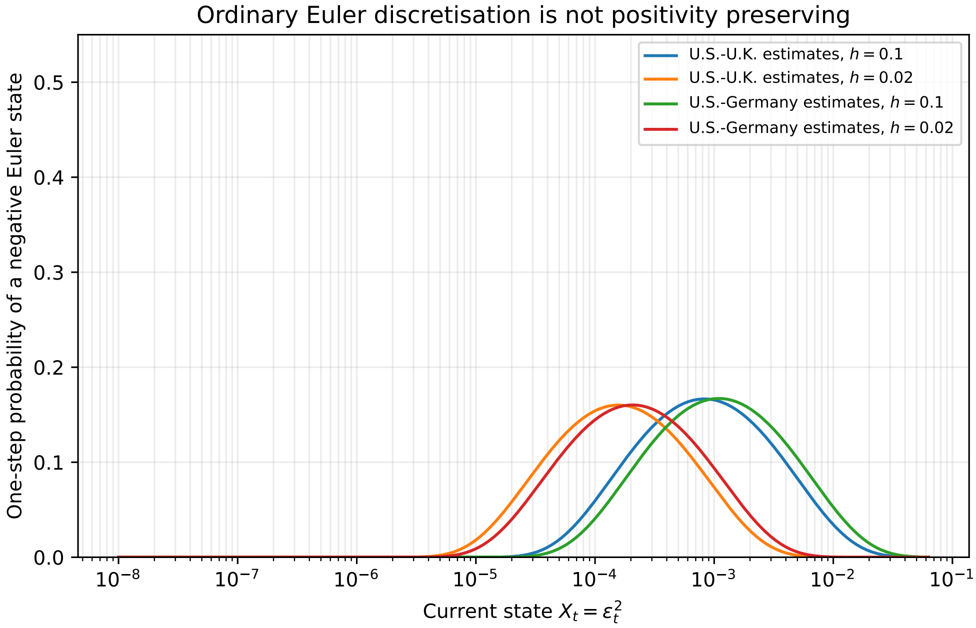

Figure 5: One-step probability that ordinary Euler discretisation of the incompleteness specification $X_t = \epsilon_t^2$ in (46) produces a negative state, using the estimates in Table 3 of Brandt and Santa-Clara (2002). The implemented estimator substitutes ϵ_t out and simulates (r, r^*, e, v) instead, so this is an illustration of the boundary mechanism rather than of the implemented chain; see Section 8.3.

estimator, not an implementation detail, because different truncation and reflection rules define different transition densities.

10.5 Practical design rule

If the unconditioned endpoint estimator is nevertheless used, M and S should be calibrated jointly. A minimum diagnostic is the effective endpoint size

$$R_{M,S} = S/M^{K/2}.$$

Increasing M without increasing $R_{M,S}$ is not an accuracy improvement. For density-level asymptotics, $R_{M,S}$ must diverge. For likelihood-level work, one should also monitor the variance and lower tail of every estimated transition density, use log-sum-exp evaluation, and verify stability after increasing both M and S along a theoretically admissible path.

11 Conclusion

The endpoint simulated-likelihood estimator has a simple but consequential dimensional structure. Its final Gaussian transition is a shrinking kernel. In K dimensions, the second moment of one simulated contribution grows as $M^{K/2}$, so the Monte Carlo variance is of order $M^{K/2}/S$. The relevant simulation size is $S/M^{K/2}$.

This leads to four conclusions about Brandt and Santa-Clara (2002), stated in decreasing order of strength. First, the joint transition-density convergence asserted in their Lemma 2 is directly false: an exact Brownian sequence satisfying the paper's own Assumptions 1 and

2, and satisfying its rate condition, makes the simulated density converge in probability to zero rather than to the positive true density. Second, because Lemmas 3 to 6 take Lemma 2 as an input, the published proofs of Theorems 1 and 2 fail. That is a statement about the proofs; we do not show that their conclusions are false, and Section B explains why the density counterexample does not transfer to the maximiser. Third, the rate condition used in those theorems is incompatible with pointwise mean-square consistency of the simulated transition density in the four-dimensional application. Fourth, and independently of all of the above, the ratios corresponding to the paper's own conditions $\sqrt{S}/M \to 0$ and $N/S^{1/4} \to 0$ are not small at the implemented design $N = 544$, $M = 10$, $S = 5{,}000$, so that design is not in a regime resembling the asymptotic ordering the theorems describe.

The corrected transition-density analysis also supplies constructive results. The density estimator has a triangular-array CLT under $S/M^{K/2} \to \infty$. If the Euler density bias is first order, the MSE-optimal discretisation is $M \asymp S^{2/(K+4)}$. The nonlinear log-likelihood scale is $M^{K/2}/S$, subject to the usual lower-tail conditions required for expectations of logarithms. These results explain why bridge proposals and other importance-sampling methods are not merely computational refinements: they address the rare-event geometry of the estimator.

The exchange-rate application raises separate questions. Its square-root states fall outside the smooth uniformly elliptic theory, although for the interest rates the Feller ratios are large enough that this is a gap in proof coverage rather than a numerical hazard. The incompleteness state is different: its Feller ratio is exactly one half for every admissible parameter value, and the identification rule drives it onto its degenerate boundary exactly. Because the estimator substitutes that state out and simulates volatility instead, the requirement becomes that a recovered square root remain real along every simulated Euler path, which ordinary Euler discretisation does not guarantee. The volatility equation is implicit when risk prices depend on volatility, and the reported U.S.–U.K. estimates give a leading coefficient of the wrong sign, so that specification admits either no positive volatility branch or two, never exactly one. The Brownian correlation matrix requires a joint positive-semidefinite restriction that entrywise bounds do not deliver. The final empirical estimator also contains a minimum-incompleteness corner rule and binding constraints that are not covered by ordinary interior SML asymptotics.

None of these points, individually or collectively, proves that the finite-sample economic conclusions are false. They show that those conclusions cannot inherit the claimed maximum-likelihood asymptotics from the mathematical argument given. The appropriate post-publication response is a corrected theorem, a clear statement of the estimator actually implemented, and a re-examination of numerical stability under dimensionally valid simulation rates or bridge-based alternatives.

A Additional proof details

A.1 The Brownian collapse for more general subcritical paths

The proof of Theorem 3.3 extends beyond $M = S$. Let $h_n \downarrow 0$ and choose a radius

$$r_n = \sqrt{ch_n \log(1/h_n)}.$$

If

$$S_n h_n^{K/2} \{\log(1/h_n)\}^{K/2} \to 0$$

and S_n grows at most polynomially in $1/h_n$, then with high probability no preterminal draw lies within r_n of the endpoint, while the maximum kernel outside that ball tends to zero for sufficiently large c. Thus the estimator again collapses to zero. The theorem uses $M = S$ because it is transparent and directly satisfies the published rate in four dimensions.

B The argmax question in a location model

This appendix records what can be proved about the simulated-likelihood maximiser in the most favourable possible model, and where the argument stops. The conclusion is negative: we do not obtain a counterexample to argmax consistency, and the manuscript makes no such claim. Two intermediate results are complete and are stated here because they clarify what the endpoint construction does to the estimation problem.

Consider

$$dY_t = \theta\, dt + dW_t, \qquad Y_t \in \mathbb{R}^K, \tag{60}$$

observed at unit intervals $t = 0, 1, \ldots, N$. The Euler scheme is exact for every M, the coefficients satisfy Assumptions 1 and 2 of Brandt and Santa-Clara (2002) exactly, and the exact maximum likelihood estimator is the mean increment $\widehat{\theta}_N = N^{-1} \sum_n \Delta Y_n$. Common random numbers across θ are used throughout, as the published implementation requires for a smooth objective: the innovations $\xi_{n,s} \sim \mathcal{N}(0, I_K)$ are drawn once and held fixed as θ varies.

Proposition B.1 (Common random numbers give an exact kernel estimator)**.** *Set $h = 1/M$, $u_n(\theta) = \Delta Y_n - \theta$ and $a_{n,s} = \sqrt{1-h}\,\xi_{n,s}$. Then for every θ,*

$$\widehat{q}_{M,S}(Y_{n+1} \mid Y_n; \theta) = \frac{1}{S} \sum_{s=1}^{S} \varphi_K(u_n(\theta) - a_{n,s}; 0, hI_K)\,. \tag{61}$$

That is, the simulated transition density is exactly a Gaussian kernel density estimate with bandwidth $\sqrt{h}$, built from a sample that does not depend on θ and evaluated at the θ-dependent point $u_n(\theta)$.

Proof. The Euler recursion for (60) is $Y^M_{m+1} = Y^M_m + \theta h + \sqrt{h}\,\varepsilon_{m+1}$, so after $M-1$ steps from $Y^M_0 = Y_n$ the preterminal state is $Z_{n,s}(\theta) = Y_n + \theta(1-h) + \sqrt{1-h}\,\xi_{n,s}$, where $\sqrt{1-h}\,\xi_{n,s}$ collects the $M-1$ independent increments. The one-step density (3) evaluated at Y_{n+1} has mean $Z_{n,s}(\theta) + \theta h$ and covariance hI_K, and

$$Y_{n+1} - Z_{n,s}(\theta) - \theta h = \Delta Y_n - \theta - \sqrt{1-h}\,\xi_{n,s} = u_n(\theta) - a_{n,s}.$$

Substituting into (5) gives (61). □

Proposition B.1 is an identity, not an approximation, and it recasts Theorem 3.3 in familiar terms: a kernel density estimate with S atoms and bandwidth $\sqrt{h}$ is uninformative when $Sh^{K/2} \to 0$.

Proposition B.2 (The maximiser is governed by a nearest-atom objective)**.** *Write $b_{n,s} = \Delta Y_n - \sqrt{1-h}\,\xi_{n,s}$ and*

$$D_N(\theta) = \sum_{n=0}^{N-1} \min_{s \le S} \|\theta - b_{n,s}\|^2.$$

Then for every θ,

$$0 \le 2h \log \widehat{\mathcal{L}}_{N,M,S}(\theta) + D_N(\theta) + NhK \log(2\pi h) \le 2Nh \log S. \tag{62}$$

Proof. Fix n and let $d_n = \min_s \|\theta - b_{n,s}\|^2$. Factoring the minimising term out of the sum in (61),

$$\frac{1}{S} \sum_s e^{-\|\theta - b_{n,s}\|^2/(2h)} = e^{-d_n/(2h)} \cdot \frac{1}{S} \sum_s e^{-(\|\theta - b_{n,s}\|^2 - d_n)/(2h)}.$$

Every exponent in the second factor is nonpositive and at least one is zero, so that factor lies in $[1/S, 1]$. Taking logarithms, adding the constant $-\frac{K}{2}\log(2\pi h)$ from the Gaussian prefactor, multiplying by $2h$ and summing over n gives (62). □

The bound is uniform in θ with slack $2Nh \log S$, which along $M = S = n$ is $2N(\log n)/n$ per unit of N and therefore negligible relative to D_N. The maximiser of the simulated likelihood is thus governed by a nearest-neighbour objective over the NS atoms $b_{n,s}$, whereas the exact log likelihood in this model is the smooth quadratic $-\frac{1}{2}\sum_n \|\theta - \Delta Y_n\|^2$ with minimiser $\widehat{\theta}_N$. The two objectives have no reason to share a maximiser. This is the precise sense in which the endpoint construction replaces the estimation problem rather than approximating it.

What is missing is the limit of D_N. A nearest-neighbour heuristic, given in the accompanying notes, suggests that $N^{-1}S^{2/K}D_N(\theta)$ converges to a multiple of $\exp\{\|\theta - \theta_0\|^2/(K-2)\}$, finite exactly when $K > 2$, which is the same threshold as Theorem 3.3. If that were established, the simulated argmax would be *consistent* for θ_0 but would converge at a rate governed by $S^{2/K}$ rather than the parametric rate, so that $\sqrt{N}(\widehat{\theta}_{N,M,S} - \widehat{\theta}_N)$ would not vanish, contradicting the conclusion of Lemma 5. Proving it requires a uniform-in-θ nearest-neighbour law for a triangular array of exchangeable, not independent, atoms with S growing; standard nearest-neighbour asymptotics are pointwise and assume independent sampling from a fixed density. We did not close that gap.

Numerical results in the accompanying notes are consistent with this picture: the simulated maximiser is consistent, its error remains a roughly constant multiple of the exact MLE error, and the scaled gap $\sqrt{N}\|\widehat{\theta}_{N,M,S} - \widehat{\theta}_N\|$ shows no downward trend. Those computations use a single seed per design and cannot reach the regime $N^4 \ll S \ll M^2$ in which the published theorem operates, so they are evidence and not proof. Accordingly the manuscript claims only what Section 7 establishes: Lemma 2 is directly refuted, and the proofs of Theorems 1 and 2 therefore fail.

C Reproducibility

The accompanying directory contains:

- `main.tex`, the manuscript source;
- `references.bib`, the bibliography;
- `code/generate_results.py`, the fully reproducible simulation and figure code;
- vector PDF and PNG versions of all figures;
- CSV and LaTeX versions of generated numerical tables.

The code uses a fixed seed, vectorised Gaussian simulation, and log-sum-exp evaluation of the endpoint estimator. No proprietary data are used.

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